Do Facial-Affect Process Data Improve Cognitive Measurement?
An Adversarial Out-of-Sample Benchmark

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Abstract. Cognitive assessments are increasingly taken in front of a camera, and commercial software converts the video into continuous scores labeled anxiety, attention, or joy. Offered to measurement as process data, these channels are said to tell an assessment something the answer record cannot. The claim is rarely tested in a form that could fail; we report such a test. The corpus is 169 college students taking a 25-item reasoning assessment, some in person and most over Zoom. We asked whether the facial channels improve out-of-sample prediction of what an examinee does next: answer correctly, take longer, ask for help, or skip. Nearly all of the evaluation was fixed in advance: models were scored on unseen examinees, "no gain" was given a calibrated width by simulation, and any gain had to beat corrupted copies of the same signals. No facial representation earned a usable gain. Four of the six primary comparisons were precise nulls, any effect smaller than a stated floor; two were inconclusive at that boundary. Richer temporal and person-specific representations did worse. One within-person comparison did pass the original screening rule, but corrupted signals with no facial content matched its gain, so it supports no facial claim either. The machinery can find a signal when one exists: allowed to see the item it was predicting, it reported a large illusory gain. What the channels did track was the recording situation, predicting in-person versus Zoom sessions better than anything cognitive, though that contrast is unresolved at the person level. The benchmark is reusable for any proposed process-data channel.

Keywords: process data; affective computing; facial expression analysis; incremental validity; predictive benchmark; equivalence testing; data leakage

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1. **Introduction**

Cognitive assessments are increasingly administered in front of a camera. Remote proctoring, video-based tutoring platforms, and webcam-equipped digital assessments all produce, as a side effect, a continuous record of the examinee's face, and a growing industry offers to convert that record into real-time streams of "anxiety," "attention," or "joy" scores (Calvo & D'Mello, 2010; Küntzler et al., 2021). The argument offered to measurement is direct: item responses are sparse and slow, whereas the face is dense and immediate, so facial-affect channels should add to the information available to an assessment, whether by flagging disengagement before errors accumulate, by personalizing pacing, or by explaining performance (Bosch et al., 2016; D'Mello & Graesser, 2012; Horvers et al., 2024; Monkaresi et al., 2017). The theoretical warrant is long-standing: emotions are entangled with learning and test taking (Pekrun et al., 2002), and the broader process-data literature has shown that logs of *how* examinees respond can carry information beyond whether they respond correctly (Fu et al., 2024; Man et al., 2022; Tang et al., 2021).

The claim on which deployment rests, however, is more specific than any of these observations, and it is tested far less often than it is asserted. It is an *incremental* claim (Sechrest, 1963): that after conditioning on everything the assessment already records about an examinee (accuracy, response time, retries, hint use, skipping, and their running histories), the facial stream still adds information about what the examinee will do next. Incremental claims are demanding because this baseline is strong; the behavioral log is itself process data of high quality, available at zero additional cost and zero additional privacy exposure. The surrounding evidence also gives reasons for doubt. Facial configurations map onto emotional states far less specifically than everyday intuition assumes (Barrett et al., 2019; Krumhuber et al., 2023), commercial classifiers disagree with each other and with human coders (Dupré et al., 2020; Stöckli et al., 2018), and in the corpus studied here, Lee and Wang (2026) found that these channels carry a highly repeatable within-session person signature whose source their design could not identify: stable morphology, expressivity, camera and lighting, and genuinely affect-relevant individual differences remain rival explanations that the design could not separate. A channel can be repeatable and still useless for measurement, or noisy and still useful; repeatability and utility are separate questions, and this article addresses the second.

The utility question has a property that complicates informal answers: in high-volume process data, weak positive findings are easy to produce. With thousands of transitions, hundreds of derived features, flexible models, and freedom in how the data are split, some specification will show a "significant incremental gain" by accident or by leakage, that is, by the unnoticed reuse of information from the future or from the test rows (Kapoor & Narayanan, 2023). The machine-learning literature has documented the pattern repeatedly: models exploit shortcuts that fail outside the distributions that support them (Geirhos et al., 2020), and apparent progress can shrink or disappear under careful evaluation (Hand, 2006). A comparable lesson has emerged within psychology itself, where elaborate affect-dynamic measures often add little to simpler summaries once baselines are taken seriously (Dejonckheere et al., 2019). If facial-affect process data are to be adopted in cognitive measurement, the evaluation should therefore be constructed so that a spurious signal would fail it.

In this article we construct such an evaluation and report its outcome for one complete, well-characterized corpus: 169 college students who took a 25-item scenario-based reasoning assessment while a commercial system scored their webcam video into nine facial-affect channels, 42 of them in person before the COVID-19 campus closure and 127 over Zoom afterward. We pose the question as prediction rather than explanation (Shmueli, 2010; Yarkoni & Westfall, 2017) and ask five things. (1) Does the facial summary block improve held-out prediction of next-item accuracy, next-item speed, or post-error help seeking and skipping, beyond the behavioral log, for people the model has never seen? (2) Do temporal organization or person-specific normalization, the representations closest to the promise of state tracking, do better? (3) Where along the ladder of representations does held-out predictive performance stop improving? (4) Do any surviving gains exceed what corrupted facial signals achieve? (5) What does the same representation know about measurement-irrelevant context, such as whether the session was run in person or over Zoom, which calibrates what a positive cognitive finding would have meant?

Because such findings are easy to come by, the evaluation was built to be difficult to pass, and its rules were fixed before any facial result was seen, with one disclosed exception that is dual-reported throughout (Section 3.5). Four commitments matter most, and Section 3 develops each in plain terms before stating its technical form. First, a frozen estimand: every comparison is scored in *usable bits per event* on held-out data, a currency in which the assessment's own log provides a yardstick. Second, a paired representation ladder inside a set of generalization regimes, so that "does the face help?" is always asked as "help *whom*, relative to *what*?" Third, equivalence regions calibrated by simulation, so that "no gain" is a bounded statement rather than a failure to reach significance (Lakens, 2017). Fourth, the design carries its own checks: a leakage register with automated verification, four families of corrupted facial signals that any genuine gain must exceed, and one deliberately leaky analysis that shows what the benchmark would report if it were broken. The full pipeline was audited three times, once by a fresh-context internal review and twice by independent external reviews, and every audit forced corrections that are part of the record (Section 3.8; Appendix F).

The answer can be stated at the outset. No facial representation survived adjudication. The facial block failed the gate in all six primary cells, with increments between -0.002 and -0.018 bits per event, indistinguishable from the penalty a same-sized block of noise columns incurs; four cells are precise nulls inside their equivalence regions and two are inconclusive at the boundary, identically under the original and the amended regions. Richer facial representations were reliably worse, and held-out predictive performance peaked at the last behavioral rung (M2), before any facial columns enter, for every outcome. The residual findings are themselves informative: the one within-person gain bounded away from zero, the only comparison to pass the original screening rule, cannot be separated from corrupted signals carrying no facial content, and the rest fall inside their equivalence regions; the deliberately leaky analysis gained a quarter of a bit per event on duration, seven times what response time itself contributes to that outcome, showing what an unguarded pipeline would have found; and, descriptively, the facial block predicted whether a session was in person or over Zoom better than it predicted any cognitive outcome.

The article makes three contributions. Empirically, it supplies an adversarially adjudicated estimate, here an equivalence-bounded null, of the incremental predictive value of commercial facial-affect channels for cognitive measurement, on a corpus whose measurement properties are unusually well documented (Lee & Wang, 2026); as far as we are aware, no prior benchmark of this kind exists for facial channels, although we have not run a systematic search to claim priority. Methodologically, it offers a reusable template (estimand, ladder, regimes, register, controls, and gate) for evaluating any proposed process-data channel, facial or otherwise. Practically, it quantifies the leakage artifact that most naturally inflates such claims, so that reviewers and buyers have a concrete demonstration to require of any positive report.

The plan of the article is as follows. Section 2 describes the corpus: the participants and their two administration modes, the facial channels, the transition frame that defines the unit of analysis, and the outcomes. Section 3 presents the benchmark:

the estimand, the representation ladder, the generalization regimes, the models, the equivalence regions, the safeguards, the decision rule, and the audit history. Section 4 reports the results, taking the five questions in turn and placing each control comparison beside the cells it adjudicates, and Section 5 discusses what the null does and does not establish, its practical implications, and the limitations of the design. Appendices A to G contain the corpus detail, the frozen design contracts, the simulation calibration, the complete result tables, the adjudication of residual signals, both audit reports, and the reproducibility interface.

2. Data: A Mixed-Mode Facial-Affect Corpus

2.1. Participants, task, and administration modes.

The corpus consists of 169 college students at a public university in the southern United States who completed a 25-item, untimed, scenario-based reasoning assessment while their face was recorded. The sessions were collected for an experiment on autonomy support and academic persistence (Wang et al., 2021); the behavioral log of the same experiment has since been analyzed in its own right (Wang et al., 2025). Items were drawn from a culture-fair reasoning pool in two versions; 62 participants took the verbal version and 107 the mathematics version. After a wrong first attempt the interface offered three options: try again, request a hint, or skip the item. Each session therefore produced, alongside the answer record, a fine-grained behavioral log of retries, hint requests, skips, and response times.

Data collection spanned the onset of the COVID-19 pandemic, and the corpus is consequently mixed-mode. Forty-two participants were tested in person between February 18 and March 13, 2020, when the campus closure halted collection; 127 were tested remotely over Zoom between September 2, 2020, and January 29, 2021. The two windows do not overlap, so the session timestamps partition the cohort exactly, and we refer to the partition as administration mode rather than by the release files' historical name, "recording wave." One discrepancy with the published record must be stated plainly: both source publications describe the collection as conducted online over Zoom (Wang et al., 2021, 2025), and the in-person subset appears only in the archival record. Among the 169 analytic participants, 162 have a survey record whose administration-setting field agrees with the timestamp partition (42 on site, 120 online); the remaining seven, all in the late window, lack a survey record and are classified as remote from their session timestamps. The composition is reported in Table 1 and, in more detail, in Appendix A. Mode is confounded with calendar time, with the capture context (the remote sessions read the participant's video window inside Zoom; the in-person capture path is not documented in the archival record), and partly with task version (7 of the 42 in-person participants took the verbal version, against 55 of the 127 remote participants), and it enters the analysis in three disciplined roles: never as a model predictor, once as a generalization split (train on one mode, test on the other), and once as a prediction *target* when we ask what the facial channels actually track (Section 4.6).

2.2. **The facial channels.** During each session a commercial facial-inference system read the webcam stream and emitted scores roughly six times per second on nine channels whose vendor source fields are *JOY*, *SAD*, *Anger*, *Surprise*, *Fear*, *Disgust*, *Contempt*, *Valence*, and *Engagement*. The analysis retains the established project-derived publication labels *Joy*, *Sad*, *Anger*, *Anxious*, *Fear*, *Disgust*, *Contempt*, *Negative*, and *Attentive*. In that crosswalk, *Surprise* becomes *Anxious*; *Engagement* becomes *Attentive* (through the historical intermediate name *Danger*); and signed *Valence* (observed range roughly -100 to 100) becomes *Negative* after absolute-value sign-folding, so valence direction is not retained (Lee & Wang, 2026). The other publication labels match their vendor fields apart from capitalization. These project-derived labels are analysis names, not vendor product names or measurements of emotional states; nothing in this article assumes that the *Anxious* channel measures anxiety, and the question we ask (does the stream add predictive information?) is well posed either way. The channels' measurement behavior on this corpus was characterized by Lee and Wang (2026): eight of the nine channels order persons stably under stress tests while *Joy* does not, and the channel scores carry a large, highly repeatable person-stable component whose source (morphology, expressivity, hardware, session context, or genuine affective style) their design could not separate. Following that evidence, the primary facial representation here uses the eight stable channels, with the nine-channel variant retained as a labeled sensitivity arm. All facial assets are consumed from the frozen releases of that study through a read-only interface that verifies a cryptographic hash on every access; no statistic of that study is re-estimated or republished here.

2.3. **The transition frame.** The unit of analysis is the *transition*: an ordered pair of adjacent item positions $(i, i+1)$ for one person, with every predictor measured at or before the end of item i and every outcome measured at item $i+1$. Figure 1 shows one participant's session laid out this way and locates the corpus around it:

panel (a) shows the item intervals with their behavioral outcomes, three of the nine channel traces, and one highlighted transition, with *time zero*, the end of item i, separating the shaded predictor history from the outcome item; panel (b) maps all 169×24 potential transitions by administration mode; and panel (c) shows how each channel's within-item stream is summarized into the three numbers per channel that the primary facial representation uses (whether the channel activated at all, the share of samples on which it was active, and the mean log magnitude of its strictly active samples). Time zero is the design's first structural safeguard against leakage: by construction, nothing measured after the end of item i can enter a prediction about item $i{+}1$.

Of the $169 \times 24 = 4{,}056$ possible transitions, 57 are unobserved because the session recording ended before the later items were reached, leaving 3,999 transitions for analysis (Table 1); 56 of the 57 missing transitions come from in-person sessions, whose recordings were more often truncated. Transitions carry their mode and task labels, and 952 of them come from in-person sessions and 3,047 from Zoom sessions.

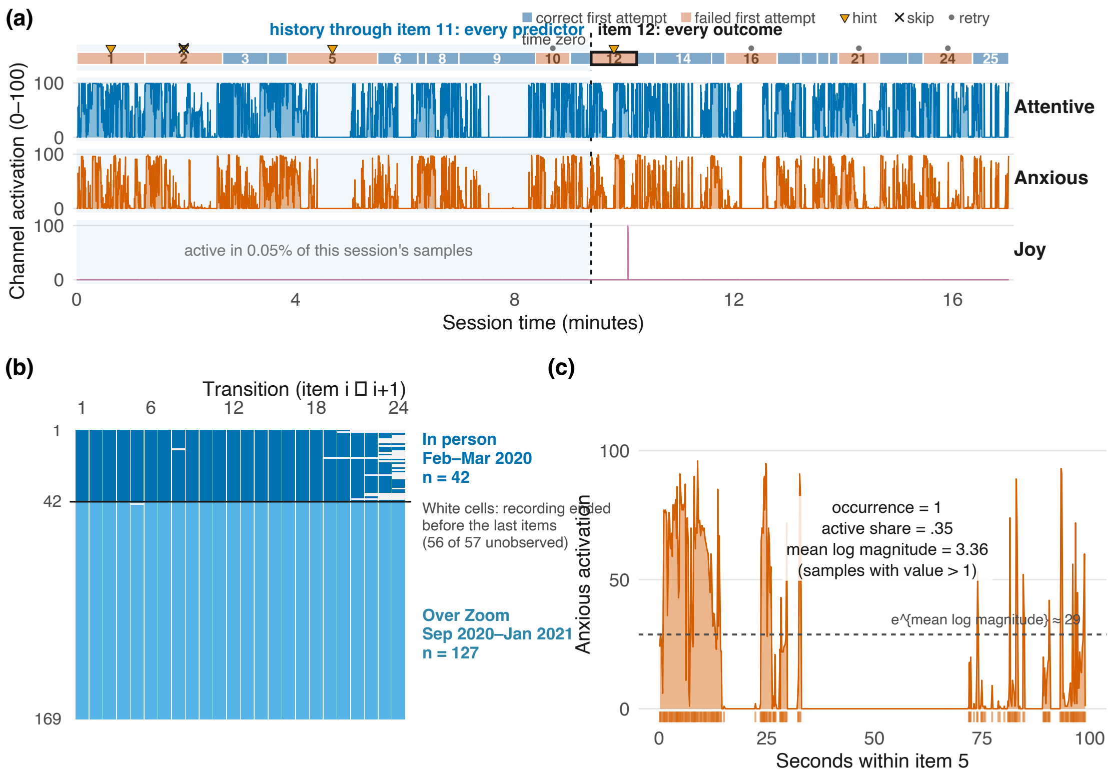
TABLE 1. The Corpus, Its Two Administration Modes, and the Prospective Transition Frame

Quantity	Value	Definition
Participants	169	completed the 25-item assessment with video
in person	42	Feb. 18 – Mar. 13, 2020; ended by campus closure
over Zoom	127	Sep. 2, 2020 – Jan. 29, 2021
Task versions	62 / 107	verbal / mathematics
Transitions $i \to i{+}1$	3,999	predictors through item i; outcomes at item $i{+}1$
in person / over Zoom	952 / 3,047	of 4,056 possible; 57 unobserved
next-item first attempt	3,999	correct on first attempt at $i{+}1$ (rate .589)
next-item log duration	3,999	log seconds on item $i{+}1$
post-error help seeking	1,645	hint use after a failed first attempt (rate .347)
post-error skipping	1,645	skip after a failed first attempt (rate .174)

Note. The behavioral log and the facial channels are consumed from the frozen releases of Lee and Wang (2026) through a hash-verified read-only interface; no statistic of that study is re-estimated or republished here. Facial predictors describe item i only and outcomes live at item $i{+}1$, so time zero separates predictor from outcome by construction. The 57 unobserved transitions arise almost entirely (56) from in-person sessions whose recordings ended before the last items. Post-error outcomes are defined only on transitions whose item-$i{+}1$ first attempt failed, which is why their event counts are smaller.

2.4. **Outcomes.** Four prospective outcomes at item $i{+}1$ were frozen from the behavioral log before any facial data were touched: *first-attempt accuracy* (binary; base rate .589), *log response duration* (continuous), and two post-error choices defined only

FIGURE 1. One Session, and the Corpus Around It



Note. (a) One illustrative remote-mode participant's complete session (mathematics version), chosen for legibility. The band shows the 25 item intervals on the session clock, colored by first-attempt outcome, with hints, skips, and retries marked above; the three traces below show the *Attentive*, *Anxious*, and *Joy* channels at the native sampling rate (roughly 6 Hz). For the highlighted transition, everything in the shaded region (behavior and face through item 11) is available as a predictor, and everything about item 12 is an outcome; the dashed line is time zero. *Joy* is shown deliberately: it is near-silent here and sparse corpus-wide, consistent with its exclusion from the primary block on the person-ordering evidence (Section 2.2). (b) The person-by-transition grid: 169 participants (rows) by 24 transitions (columns), 3,999 of 4,056 cells observed. Rows are grouped by administration mode; 56 of the 57 unobserved transitions belong to in-person sessions whose recordings ended before the last items. (c) Construction of the facial summary block from one item's stream: per channel, an occurrence indicator, the active share of samples, and the mean log magnitude of the strictly active samples (those above the activation threshold of 1); the magnitude summary is a conditional mean on the log scale, not a peak. The three summaries give 24 predictors over the eight primary channels.

when the first attempt fails, *help seeking* (hint use, given no skip; rate .347 among the 1,645 eligible transitions) and *skipping* (rate .174). Skip takes precedence in the taxonomy, making the post-error categories mutually exclusive. These four were chosen because they are the behaviors that deployment arguments promise to anticipate:

whether the examinee will succeed, how long the next item will take, and whether an error will be followed by seeking help or by disengagement.

A second, retrospective use case scores held-out items from an observed block: after observing a person's first k item positions (primary $k = 12$; sensitivity $k \in \{5, 8, 15, 18\}$), predict first-attempt accuracy and log duration for each remaining position. This corresponds to the practical question of whether, given a partial record that includes video, the rest of the test is better predicted with the face than without it.

3. A Benchmark That Could Fail

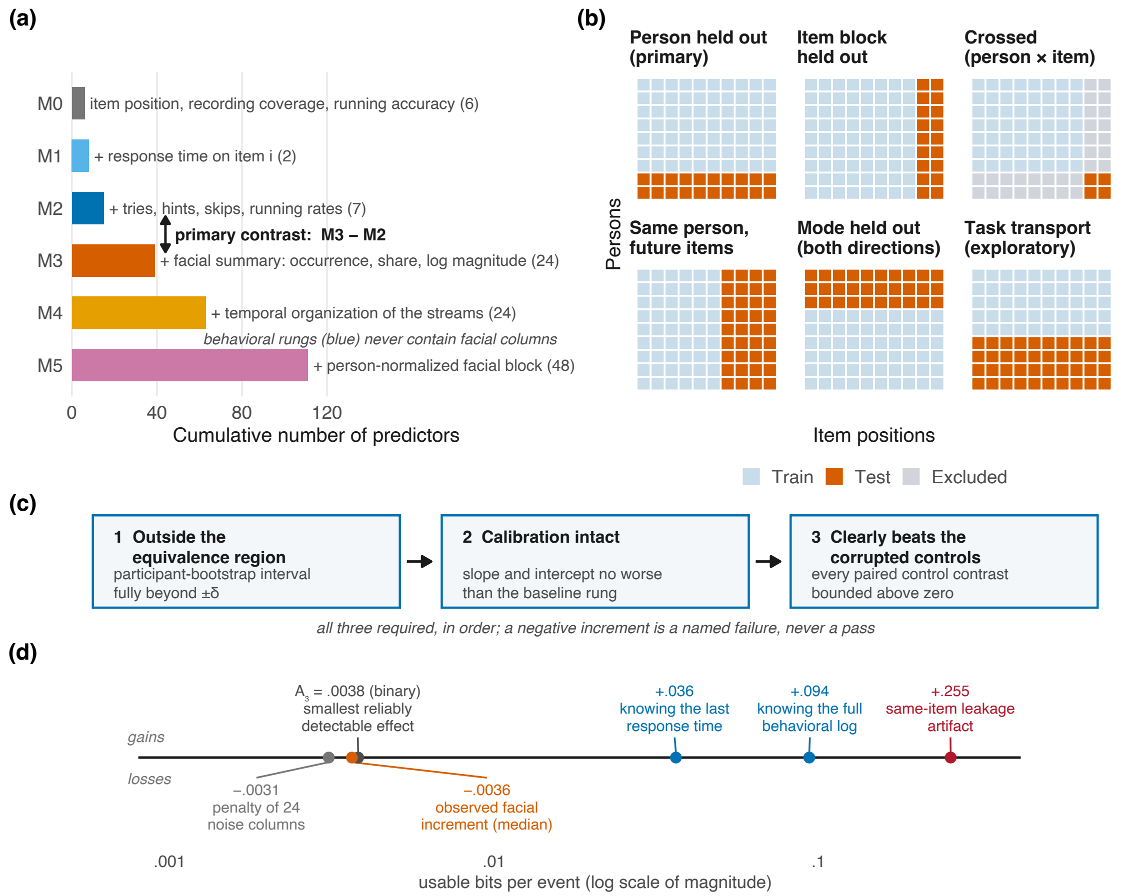
This section presents the evaluation in the order a reader needs it: what is being measured (the estimand), what is being compared (the ladder), for whom the predictions are made (the regimes), how the models are fitted and their uncertainty summarized, what "no gain" means (the equivalence regions), what protects the comparison from manufactured gains (the safeguards), and how a verdict is reached (the decision rule). Two closing subsections record how the design was frozen and audited, and what is openly shared. The frozen contracts themselves, with timestamps and identifiers, are reproduced in Appendix B; readers who want every arm and number will find them in Appendices C to E. Figure 2 collects the design in one place.

3.1. The estimand: usable bits per event. Every comparison in this article is a paired difference in out-of-sample predictive score between two representations of the same transitions. Each model emits, for every held-out transition, a predictive distribution for the outcome, and that prediction is scored with the logarithmic score in base 2, the strictly proper rule that rewards a forecast according to the probability it assigned to what actually happened (Gneiting & Raftery, 2007). For a binary outcome y_j with predicted probability $\hat{p}_j$, the score of row j is $\log_2 \hat{p}_j$ if $y_j = 1$ and $\log_2(1 - \hat{p}_j)$ otherwise; for log duration, the score is the log of the Gaussian predictive density, $\log_2 \phi\{(y_j - \hat{\mu}_j)/\hat{\sigma}_{f(j)}\} - \log_2 \hat{\sigma}_{f(j)}$, where $\hat{\sigma}_{f(j)}$ is the residual standard deviation estimated on the training portion of row j's own fold. The estimand for a comparison of a higher rung against a lower rung is the mean paired difference

$$\Delta U \;=\; \frac{1}{n}\sum_{j=1}^{n}\Big(s_j^{\text{hi}} - s_j^{\text{lo}}\Big), \tag{1}$$

where s_j is the row-j log score, in *usable bits per event*.

FIGURE 2. The Benchmark: Ladder, Regimes, Gate, and the Scale of the Units



Note. (a) The representation ladder. Each rung adds one block of predictors to everything below it, so adjacent rungs differ by exactly one block and every contrast is paired; the primary contrast is M3 − M2, the increment of the facial summary over the full behavioral log. (b) Train/test geometry of the six generalization regimes on the person × item-position grid. (c) The three gate conditions, applied in order; all three are required. (d) Calibrated magnitudes in usable bits per event (log scale of absolute size; sign and role are categorical, gains above the line, losses below; quantities from different outcome families share the axis only as magnitudes). The design resolution A_3 applies to the binary outcomes only. The observed facial increment (vermillion) is indistinguishable from the noise-column penalty.

Two properties of this currency do the interpretive work. First, it is signed and cost-aware: a block of predictors that carries no information does not leave ΔU at zero but drags it slightly negative, because the model wastes fit on noise; Section 3.5 measures that penalty directly, so that the reader can tell "no information" from "small information." Second, it is anchored: the same pipeline scores the behavioral rungs, so the value of, say, knowing the previous response time (+.036 bits per event for predicting the next item's duration) is available as a yardstick for judging any facial increment (Figure 2d). We emphasize that ΔU is deliberately model-class-relative: it measures what a disciplined analyst can extract with the models described below, not the mutual information the face "contains," and no result in this article speaks to representations outside that class.

3.2. **The representation ladder.** Rather than comparing "facial model" against "no facial model" in a single jump, the analysis climbs a ladder of six nested representations (Table 2), each adding one interpretable block to everything below it. M0 carries design information only: the item position and its running count, task version, running accuracy, and two recording-coverage terms included so that the mere amount of recording cannot masquerade as facial content (six columns). M1 adds the response time on item i and its running mean (two columns); M2 adds the rest of the behavioral log (tries, hints, skips, and running rates). M3 adds the facial summary block of Figure 1c: per channel, an occurrence indicator, the active share, and the mean log magnitude of the strictly active samples on item i, 24 columns over the eight primary channels. (The frozen contract described the magnitude summary as a peak; the implemented feature is this conditional mean, and the correction is recorded in the amendment register, Appendix B.) M4 adds temporal-organization diagnostics of the item-i stream (lag-1 autocorrelation, an effective-sample-size ratio, and a variability flag per channel), which enter only if they pass an outcome-free eligibility test described in Appendix B. M5 re-expresses the facial block as a training-fold person mean plus the within-person deviation, the representation closest to the idea that departures from a person's own habitual level are what carry state information.

The ladder design has two consequences worth stating plainly. Because every rung contains all lower rungs, the contrast between adjacent rungs isolates the added block, and because all rungs within a cell share identical rows, folds, and seeds, each contrast is paired at the transition level. And because the behavioral rungs sit below every facial rung, the facial blocks are always asked the deployment-relevant question: what do you add *after* the log the assessment keeps anyway?

One scope note on M5 is required. Under person transport a held-out person's mean cannot be estimated from their own history, so their mean block is the training grand mean and the deviation block is a recoding of the raw values; M5 in the primary regime is therefore a redundant reparameterization stress test of the facial block rather than a personalization test. The registered person-specific model (mean, deviation, and

TABLE 2. The Representation Ladder and the Question Each Rung Asks

Rung	Adds	k	A gain over the rung below means
M0	position, coverage, running accuracy	6	(baseline: design alone)
M1	response time on item i	8	pacing carries information
M2	tries, hints, skips, running rates	15	the full log carries more
M3	facial summary: occurrence, share, magnitude	39	channel summaries add signal
M4	temporal organization of the streams	63	temporal summaries add more
M5	person-normalized facial block	111	deviations from habit matter

Note. k = cumulative predictor count. Every predictor is measured at or before the end of item i. The magnitude summary is the conditional mean log magnitude of strictly active samples (AMD-5 corrects the contract's original 'peak' wording; nothing is refit). The facial block covers the eight channels whose person ordering survived the stress tests of Lee and Wang (2026); the ninth (Joy) enters only a labeled sensitivity arm. M5 person means are estimated inside each training fold; a held-out person receives the training grand mean, so under person transport M5 is a redundant reparameterization stress test, and the registered mean-by-deviation interaction model is evaluated in the within-person regime instead (Section 4.5). Every rung within a cell is fitted on identical rows and folds, so contrasts are paired.

their interaction) is evaluated where personalization is actually possible, in the within-person regime, where a test person's own earlier items supply a real normalization history (Section 4.5).

3.3. **Generalization regimes: help whom?** A claim that facial channels "help" is incomplete until it says for whom the prediction is made, and the six regimes of Figure 2b keep the possible claims separate. *Person held out* is primary: the model never sees the test examinee, which is the situation implied by deploying a trained system on new test takers (5 folds by person, 50 repetitions). *Item-block held out* and *crossed* (train rows share neither person nor item with the test cell) probe transport to new items. *Mode held out* trains on the in-person sessions and tests on the Zoom sessions and the reverse, and *task transport* does the same across task versions; both are context-shift probes, the second exploratory by design. Finally, *same-person-future* trains on every person's items $1, \ldots, c$ and tests on their items $c{+}1, \ldots, 25$ (cut positions $c \in \{8, 10, 12, 14, 16\}$ crossed with four Monte Carlo seeds). This is the regime in which a personalized state signal would appear even if nothing transported across persons, and it is also the regime most prone to flattering artifacts, which is why it carries its own control battery (Section 4.5). The retrospective block-scoring use case of Section 2.4 is evaluated in its own, seventh regime built through the same machinery.

3.4. **Models, folds, and two kinds of uncertainty.** All models are ridge-penalized generalized linear models (Friedman et al., 2010): logistic for the binary outcomes, Gaussian for log duration. The penalty is chosen by five-fold cross-validation *inside* the training fold only (Stone, 1974), standardization uses training statistics only, and zero-variance screening operates on the training fold only, so no choice anywhere in the fitting path sees a test row. We chose this model class deliberately: it is strong enough to extract linear and additive structure from blocks of this size, transparent enough that its complexity cost can be calibrated exactly (Section 3.5), and it matches what deployment pipelines for tabular summaries typically use. A registered gradient-boosting benchmark was planned and not executed; the limitation this creates is stated in Section 5.5.

Uncertainty is reported at two levels with distinct roles, and the distinction matters for how the tables are read. The range of a statistic across repeated fold assignments (2.5th to 97.5th percentile across repetitions) describes *split stability* only: how much the number moves when the same people are partitioned differently. It is never treated as a confidence interval. The inferential object is a participant-cluster bootstrap: persons are resampled as whole clusters ($n = 169$; 2,000 resamples), and each resample statistic is the event-weighted mean, the sum of per-person bit contributions divided by the sum of their event counts, so that the interval estimates exactly the registered bits-per-event functional of Equation 1 rather than an equal-person average of person means (the latter is reported only as a labeled sensitivity). Verdicts are judged on this interval.

3.5. **Equivalence regions calibrated by simulation.** Because the hypothesis of interest is a null hypothesis, "no gain" had to be given a magnitude in advance (Lakens, 2017): an interval $[-\delta, +\delta]$ within which an increment is declared too small to matter. Two blind simulations, run on the real feature matrix with only the facial block manipulated, calibrated what this design can and cannot see; both operate on exactly the 24-column eight-channel block the primary analysis uses. A *null false-gain* distribution replaced the facial block with a person-shuffled copy within item, 100 times per outcome family, and established how ΔU behaves when the added block is pure noise: negative in 99 of 100 first-attempt replicates, mean -0.0031 bits, which is the complexity penalty of 24 uninformative columns and the correct reference point for the observed contrasts. An *injection–recovery* experiment coupled a synthetic effect of known size to the outcome and measured the smallest effect the full pipeline recovers reliably: $A_3 = 0.0038$ bits per event was detected in at least 90% of replicates. A_3 is calibrated on the binary first-attempt family and floors the binary outcomes only; the Gaussian outcomes' regions come from their much larger behavioral yardsticks, and no recovery resolution is claimed for them. Verdicts about binary contrasts smaller than A_3 are claims this design cannot make. The equivalence half-width for each outcome is $\delta = \max(0.5 \times \text{the outcome's largest behavioral yardstick layer, } A_3)$: an increment is "practically zero" if it is small against what the behavioral log itself contributes and within the design's demonstrated resolution. The yardstick component was fixed before any facial result existed; the A_3 floor was confirmed after provisional gate results had been seen, a timing that deviates from our own freeze rule. The deviation is disclosed in the amendment register (Appendix B), and every verdict in this article is reported under both the pre-amendment and the confirmed regions; the two sets of verdicts are identical.

3.6. **Safeguards: the leakage register and the control battery.** Three layers of protection run alongside the analysis, in increasing order of adversarial intent.

First, a *leakage register*. Every predictor used anywhere in the study carries four mandatory metadata fields (information time, permitted use cases, leakage class, fold scope), and automated checks plus a runtime assertion on every fitted model's column set run before any fit; an analysis that fails them is invalid regardless of its result (Kapoor & Narayanan, 2023).

Second, four *corrupted-facial control arms*, each of which destroys a stated aspect of facial content while preserving a stated aspect of structure, so that any genuine facial gain must exceed what its own corrupted twins achieve. *Circular shift* rotates a person's facial blocks across their items: the person's overall facial profile survives, but which face went with which item is destroyed. *Misalignment* attaches the facial block of item $i-1$: content survives, timing is wrong. *Identity swap* gives each row a different person's facial block from the same item and the same response-duration tertile, with a guaranteed derangement (no row keeps its own face): item-level and pacing-level structure survive, personal facial content is gone. *Marginal-matched pseudo-channels* permute each facial column independently: every column keeps its marginal distribution, but within-row coherence is destroyed. If the observed facial block cannot outpredict these four, whatever it carries is not the aligned, person-specific, item-specific content that a state-tracking story requires.

Third, one *deliberately leaky positive control*. The same-item probe commits, on purpose, the alignment error that the transition frame excludes: it scores item $i+1$'s outcomes with item $i+1$'s own facial block, information that is only available after the outcome has happened. Its role is to demonstrate that the pipeline reports large gains the moment a leakage path is opened, so that the null results elsewhere cannot be attributed to a benchmark that shrinks everything toward zero, and to quantify the artifact an unguarded analysis would report as discovery.

All controls run in the same release as the observed arm, on matched rows; the retrospective cells are adjudicated against retrospective corruptions built through the same observed-block aggregator, not borrowed prospective ones; and no control can be selected or dropped after results are seen. A mean-only versus deviation-only ablation additionally decomposes any facial contribution into person-stable and within-person components.

3.7. **The decision rule.** A cell passes the gate only if three conditions hold in order (Figure 2c): (1) the participant-bootstrap interval lies entirely outside the equivalence region; (2) calibration does not deteriorate (the added rung's calibration slope no farther from 1 than the base rung's, and its intercept no farther from 0, within registered tolerances); and (3) the observed inferential lower bound exceeds the largest corrupted-control level in the same regime, with a missing battery yielding the named verdict *controls not evaluated* rather than a silent pass. The registered reporting minimum names the form of that comparison: a *paired* adversarial control contrast, the observed gain minus the corrupted-arm gain on the same resampled participants, so that control-side uncertainty propagates rather than being collapsed to a point. Two scalar summaries of the control arms are also defensible and are reported as sensitivities: the median across repetitions, which the original implementation used, and the event-pooled point, which is the same functional as the observed bound. The distinction matters in exactly one cell of this study, and we say so where it arises (Section 4.5). No significance filter and no preferred direction enter the rules beyond the definition that a useful gain is positive; a negative increment is routed to its own named failure mode rather than suppressed, and the remaining failure verdicts are pre-named (precise null; inconclusive; negative-outside; ranking-only; artifact-not-separated; controls-not-evaluated). A transport claim additionally requires at least five repetitions behind the interval and survival of the same control battery in that regime, so single-split point estimates license no claim. In the within-person regime the same event-weighted participant-cluster bootstrap is the inferential object, and condition 3 is adjudicated on the paired contrasts just described; both were supplied in an addendum release after the second external review found that Phase 12 had judged these cells on repetition quantiles (Appendix B, AMD-5). The repetition ranges are reported as stability diagnostics.

3.8. **Freeze discipline, amendments, and two audits.** Everything that determines what counts as evidence in this article (outcome definitions, the ladder, fold assignments, the estimand and scoring rules, the leakage register, the control battery, the equivalence regions, and the gate) was frozen in machine-readable, timestamped configuration files before the first facial predictor entered any model. Later changes were permitted only through a disclosed amendment register; five amendments have occurred, each recorded with its trigger, timing, and direction of risk, and the one that followed a look at provisional results (the A_3 floor, AMD-2) is handled by dual reporting throughout. The full pipeline was audited three times: first by a fresh-context internal review, an AI system of the same family that assisted the analysis, given no prior exposure to the project and a written brief to refute our results; then by an independent external review of the whole research program; and, after this revision was assembled, by a second independent external review of the present build, whose corrections (recorded as AMD-5) include the registered-interval re-adjudication of the within-person cells reported in Section 4.5. The audits found real defects, including a fold-hygiene leak in the person-normalization transform, two unreported positive within-person cells, a calibration comparator on the wrong feature scope, an uncertainty object attached to the wrong functional, and controls weaker than their register descriptions. Every finding was repaired and the affected phases rerun; every number and rule reported here reflects the corrected record. Appendix F reproduces the audit reports, the dispositions, and what changed. We describe this history because it altered results that we would otherwise have reported incorrectly, and because the corrected record is the strongest argument we can offer that the remaining numbers mean what they say.

3.9. **Transparency and openness.** We report how the sample size was determined (inherited in full from the source corpus; no case was added or excluded by this study), all data exclusions (the 57 structurally unobserved transitions), all manipulations (none; the study is observational), and all measures in the analysis. The study was not preregistered on an external registry; the freeze-and-amend discipline described above, with timestamped contracts shipped in the replication package, is offered in its place. Analysis software is listed in the end matter. From the frozen releases, the package rebuilds all 20 tables and six of seven figures and audits 25 selected printed values under an explicit typed contract; Figure 1 is withheld at participant grain.

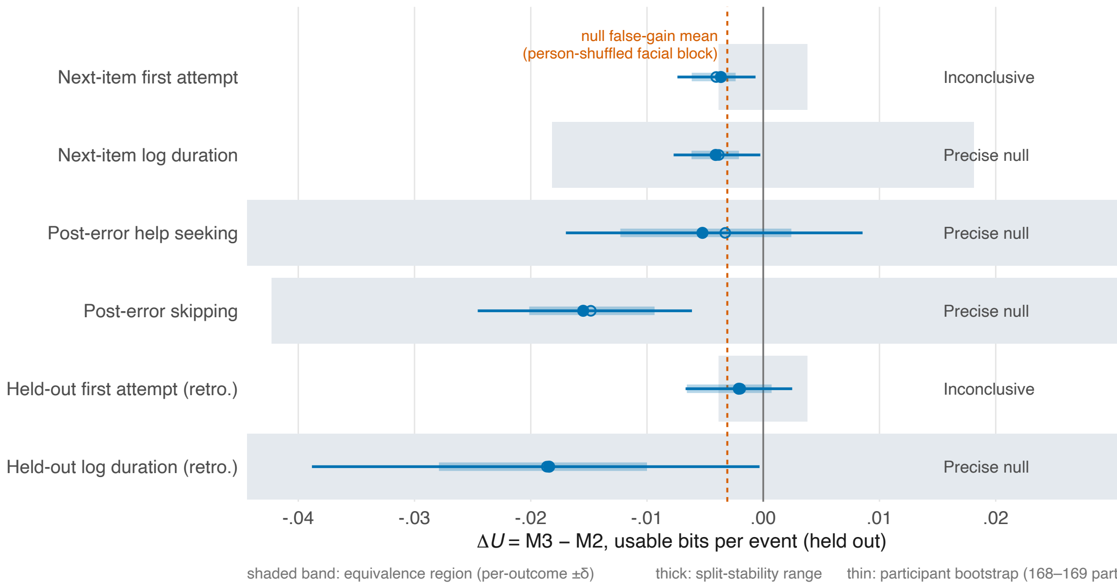
4. Results

4.1. **The behavioral yardstick.** The baseline ladder establishes what genuine information gains look like in these units, produced by the same pipeline, folds, and scoring as every facial contrast. Under person transport, adding item-i response time (M1) yields +0.036 bits per event for predicting next-item duration, and adding the remaining behavioral log (M2) yields +0.094 and +0.085 bits for post-error help seeking and skipping. These increments are large against everything the facial block will turn out to contribute (Figure 2d gives the scale). The corpus therefore contains extractable process information in quantity; the question is whether the face adds to it.

4.2. **The facial block adds nothing that survives adjudication.** Table 3 and Figure 3 give the primary result. In all six primary cells (four prospective outcomes under person transport and two retrospective block-scoring outcomes) the M3 − M2 contrast is negative: the split medians are −0.0036 (next-item first attempt), −0.0041 (next-item duration), −0.0052 (help seeking), −0.0155 (skipping), −0.0021, and −0.0184 (the two retrospective cells). Adjudication uses the event-weighted participant-cluster intervals: −0.0041 $[-0.0074, -0.0007]$ bits for first attempt, −0.0038 $[-0.0077, -0.0003]$ for duration, −0.0033 $[-0.0170, +0.0085]$ for help seeking, −0.0148 $[-0.0246, -0.0061]$ for skipping, and −0.0020 $[-0.0067, +0.0025]$ and −0.0186 $[-0.0388, -0.0003]$ retrospectively. Under the confirmed equivalence regions, four cells are *precise nulls* (their intervals lie entirely inside $[-\delta, +\delta]$) and the two first-attempt cells are *inconclusive* (their intervals straddle the boundary $-\delta = -0.0038$); the verdicts are identical under the stricter pre-amendment regions, so 0 of 6 cells pass either way. No cell reaches the control condition either: every inferential lower bound lies below the best same-regime corrupted-control median, including in the retrospective cells, which carry their own battery (all eight retrospective corrupted arms are themselves negative, medians −0.0025 to −0.0312).

The magnitude of these contrasts is as informative as their sign. The null false-gain distribution of Section 3.5, obtained by feeding the same pipeline a person-shuffled facial block of the same 24 columns, averages −0.0031 bits for first attempt; the observed −0.0036 is indistinguishable from it (the dashed reference in Figure 3).

Figure 3. Primary Cells: M3 − M2 Against the Confirmed Equivalence Regions



Note. Filled points are split medians (thick bars: split-stability ranges); open circles and thin bars are the event-weighted participant-cluster bootstrap, the inferential object on which gate condition 1 is judged (168–169 participants, depending on the cell's eligible events). Shaded bands are each outcome's confirmed equivalence region; the dashed line marks the mean of the null false-gain distribution, which is calibrated on the first-attempt family and shown across rows for scale only. Four cells are precise nulls and the two first-attempt cells are inconclusive at the boundary, identically under both equivalence regions; no cell passes the gate.

Table 3. Primary Adjudication: M3 − M2 in the Six Primary Cells

Outcome	Mdn	Split range (stability)	Bootstrap (inferential)	δ	Verdict: Confirmed	Verdict: Pre-amend.
Next-item first attempt	−.0036	[-.0062, -.0024]	[-.0074, -.0007]	.0038	Inconclusive	Inconclusive
Next-item log duration	−.0041	[-.0062, -.0021]	[-.0077, -.0003]	.0181	Precise null	Precise null
Post-error help seeking	−.0052	[-.0123, .0024]	[-.0170, .0085]	.0468	Precise null	Precise null
Post-error skipping	−.0155	[-.0201, -.0093]	[-.0246, -.0061]	.0423	Precise null	Precise null
Held-out first attempt (retro.)	−.0021	[-.0066, .0007]	[-.0067, .0025]	.0038	Inconclusive	Inconclusive
Held-out log duration (retro.)	−.0184	[-.0279, -.0100]	[-.0388, -.0003]	.0493	Precise null	Precise null

Note. Usable bits per event; negative values mean that the facial rung predicts held-out events worse than the behavioral baseline once its complexity cost is paid. The split range describes variability across repeated fold assignments and is released as a stability diagnostic only; gate condition 1 is judged on the event-weighted participant-cluster bootstrap interval (168–169 persons, depending on the cell). δ = confirmed equivalence half-width, the larger of the outcome-specific behavioral yardstick and the design resolution $A_3 = 0.0038$; the last column re-judges each cell under the pre-amendment region, which is stricter for the two first-attempt outcomes and identical elsewhere. No cell passes the gate under either region.

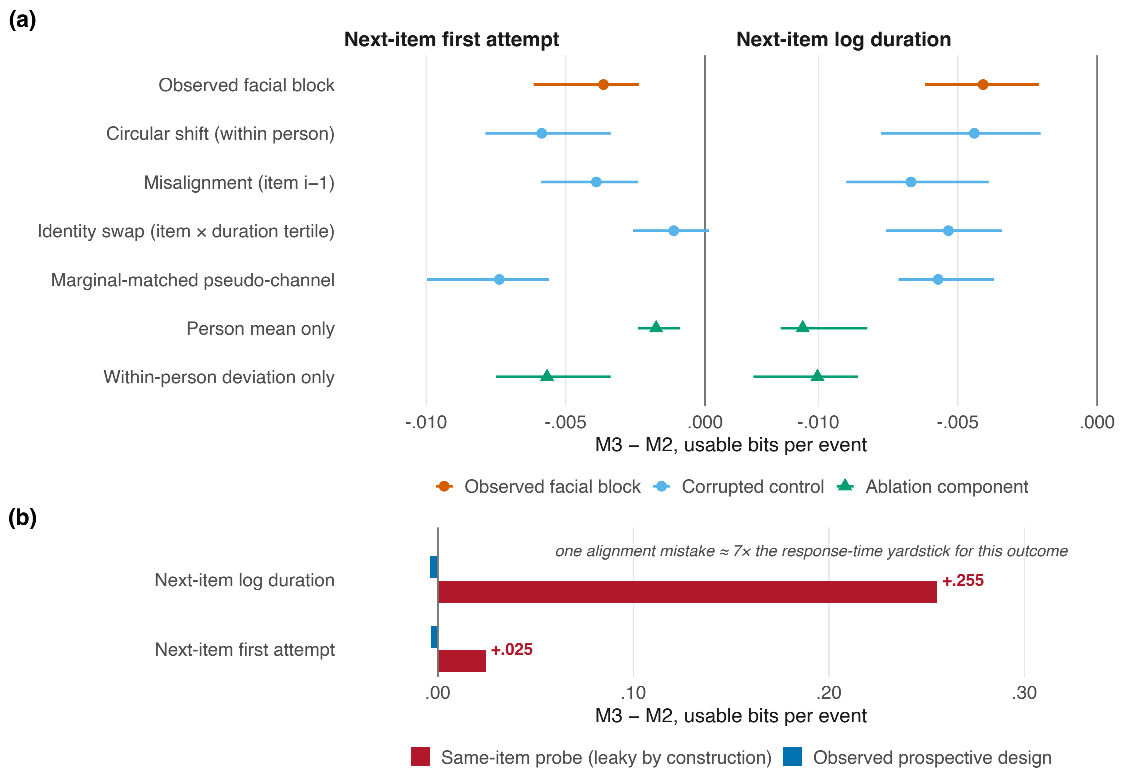
To the resolution this design can certify, the facial summary block behaves like 24 columns of structured noise paying their complexity penalty.

4.3. **Corrupted signals do as well; a leaky analysis does far better.** Figure 4a overlays the observed contrast on the four corrupted-facial arms and the mean/deviation ablation for the two headline outcomes. The observed block and its corrupted twins occupy the same negative band, and the person-mean-only and deviation-only components are individually negative as well, so there is no positive facial contribution for the decomposition to apportion. The comparison that anchors the design is panel (b): scoring item $i{+}1$'s outcomes with item $i{+}1$'s own facial block, the same-item alignment that a processing pipeline can commit in a single misplaced join, "gains" +0.255 bits per event on duration and +0.025 on first attempt. One exposure-time pathway is directly visible: facial-activation occurrence rises with recording length (within-item $r = .49$ for the primary block), so a same-item block partly encodes the outcome it pretends to predict; the association illustrates the pathway without decomposing the artifact. The benchmark therefore detects information, including artifactual information, whenever information is present; in the correctly aligned design nothing exceeded the calibrated regions or the corrupted controls. The artifact is large on any family-valid scale: the first-attempt probe (+0.025) is roughly six times the binary design resolution A_3, and the duration probe (+0.255) is seven times what response time itself contributes to that outcome and fourteen times its equivalence half-width. This is why Section 5.3 argues that any positive report of facial gains should be accompanied by this probe.

4.4. **Richer facial representations do worse.** If the facial summary discarded usable structure, the temporal (M4) or person-normalized (M5) representations should recover it; both instead performed worse. Under person transport, M4 − M3 is negative for all four outcomes (medians −0.0016 to −0.0053), and M5 − M4, with person means estimated strictly inside training folds, is negative for all four as well (medians −0.0024 to −0.0184). Figure 5 shows the resulting frontier: for every outcome, held-out codelength (the negative mean log score, so lower is better) is minimized at M2 and rises through the facial rungs on the frontier (M3 and M4). Under a description-length reading (Grünwald, 2007), the shortest attainable description of these examinees' future behavior uses no facial predictors.

4.5. **Within-person gains exist and fail their controls.** The same-person-future regime (train on a person's early items, predict their later items) produced the study's

FIGURE 4. Adversarial Controls, Ablation, and the Positive Control



Note. (a) Observed M3 − M2 versus the corrupted-facial controls and the person-mean/deviation ablation, person-held-out regime, with split-stability ranges. (b) The deliberately leaky same-item probe against the observed prospective design, same units. The probe commits the alignment error the transition frame excludes; its duration gain is seven times what response time itself adds for that outcome, and its first-attempt gain is roughly six times the binary design resolution A_3.

only consistently positive contrasts. Their reporting history is part of the record: the first audit surfaced them (they had been computed but not reported), and the second external review then found that Phase 12 had adjudicated them on repetition quantiles where the uncertainty contract requires the participant-cluster bootstrap. The registered interval was therefore supplied in an addendum release whose refits reproduce every released repetition exactly (AMD-5, Appendix B), and Table 4 and Figure 6 report the corrected adjudication.

Under the registered event-weighted functional the first-attempt gain is genuinely positive: $+0.0262$ bits per event, interval $[+0.0197, +0.0333]$, outside its equivalence region, with calibration intact. Conditions 1 and 2 are met, and the verdict turns on the control comparison. In the form the reporting minimum registers, the paired contrast, the addendum release refits each corrupted arm under the identical

Figure 5. Held-Out Codelength Frontier

Note. Median held-out codelength (negative mean $\log_2$ score, so lower is better) by cumulative predictor count. The circled minimum lies at M2, the full behavioral log with no facial columns, for every outcome.

seeds and resamples participants jointly, so that observed and control gains rise and fall on the same people. Two of the four paired contrasts exclude zero in the observed direction (circular shift, $+0.0066$ $[+0.0017, +0.0118]$; misalignment, $+0.0289$ $[+0.0221, +0.0356]$) and two do not: the deranged identity swap (another person's facial block, same item and response-duration tertile) trails the observed gain by only $+0.0049$ $[-0.0003, +0.0099]$, and the pseudo-channel arm is statistically indistinguishable from the observed block, -0.0023 $[-0.0068, +0.0026]$, with a point estimate on the wrong side. Condition 3 requires separation from every control, so the cell is *artifact-not-separated*.

The two scalar summaries of the control arms disagree with each other here, in the only cell of the study where the choice matters, and both are released. Against the repetition medians (the level the original implementation used, the largest being $+0.0052$) the lower bound clears every control and the cell passes all three conditions as historically implemented. Against the event-pooled points, the functional of the observed bound itself, it clears none, because the pseudo-channel arm reaches $+0.0285$ and the identity swap $+0.0213$. Every one of these adjudications postdates the frozen release, so none is privileged by registration; we report all three and rest the verdict on the paired contrast, which is the comparison the registered reporting minimum names and the only one that propagates control-side uncertainty. (The same scalar dual read applied to the six primary cells changes none of their verdicts, since every primary bound already fails condition 1.)

What the control pattern identifies is limited, and we state the limit. The identity swap preserves item identity and response-duration bin by construction, and it retains most of the observed gain; that much points to item-level information, carried by any block indexed to the same items, rather than to the person's own facial content. The pseudo-channel arm supports no mechanism claim: it is one fixed per-column permutation of the source cells, approximately marginal-matched on the evaluated transitions, and it preserves neither item structure nor within-row coherence, so why it matches the observed gain is not identified by this design. What its performance does establish is the negative fact the gate needed: gains of this size arise in this regime from blocks carrying no usable facial content. (The event-weighted observed value exceeds the repetition median, $+0.0111$, because the functional weights the long-horizon cuts, where position extrapolation is worst, most heavily; the repetition ranges remain in Table 4 as stability diagnostics.) For duration, the registered interval is $[+0.0028, +0.0166]$, inside the ± 0.0181 region: a precise null under every reading, and the paired contrasts run against it, with two corrupted arms exceeding the observed gain outright (misalignment by -0.0443 $[-0.0593, -0.0277]$, circular shift by -0.0088 $[-0.0178, -0.0009]$). No within-person utility claim is licensed on either count.

Two smaller residuals complete the picture. The M4 − M3 duration gain ($+0.0044$, a quarter of that outcome's δ) collapses to zero when the temporal block is rotated within person ($+0.0006$) or swapped across persons ($+0.0002$); the trace is therefore content-dependent rather than a structural artifact, but it sits well inside the equivalence region, and in the corrected battery its repetition interval includes zero (Appendix E documents both estimates of this cell and their provenance). And the registered person-specific model (person mean, deviation, and their interaction), evaluated in the only regime where a test person's own history is available, yields a median of $+0.0008$ bits on first attempt, a fifth of that outcome's δ (repetition interval $+0.0001$ to $+0.0232$), and is negative for duration (-0.0069). Even where personalization is possible, no usable gain appears. We report these traces as boundaries of the null rather than as signals.

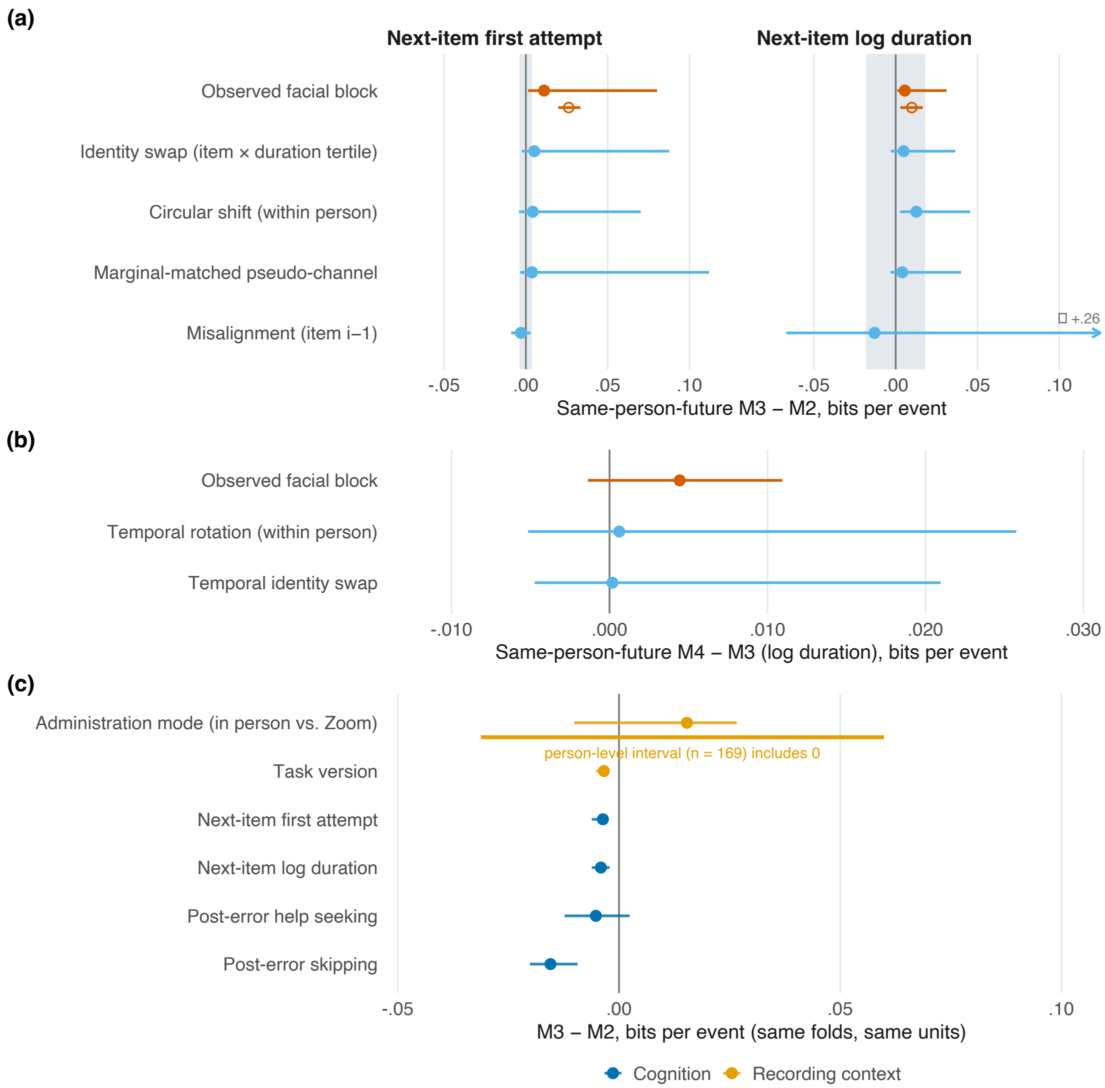
Table 4. Within-Person Residual Signals and Their Adjudication (Same-Person-Future Regime)

Contrast, cell / arm	Point	Registered interval	Stability range	δ	Ctrl. med.	Ctrl. pool.	Judgment
M3 − M2, first attempt	.0262	[.0197, .0333]	[.0014, .0802]	.0038	.0052	.0285	Hist. pass / not sep.
paired, vs. circular shift	.0066	[.0017, .0118]	—	—	—	—	excludes zero
paired, vs. misalignment +1	.0289	[.0221, .0356]	—	—	—	—	excludes zero
paired, vs. identity swap	.0049	[-.0003, .0099]	—	—	—	—	includes zero
paired, vs. pseudo-channel	-.0023	[-.0068, .0026]	—	—	—	—	includes zero
M3 − M2, log duration	.0098	[.0028, .0166]	[.0009, .0311]	.0181	.0126	.0542	Precise null
M4 − M3 dur., observed block	.0044	—	[-.0014, .0109]	—	—	—	below δ
M4 − M3 dur., temporal rot.	.0006	—	[-.0052, .0258]	—	—	—	collapses
M4 − M3 dur., temporal swap	.0002	—	[-.0047, .0210]	—	—	—	collapses

Note. Top: the two positive M3 − M2 cells, adjudicated on the registered event-weighted participant-cluster bootstrap supplied by the AMD-5 addendum release. Condition 3 rests on the paired adversarial contrasts (indented rows): observed minus corrupted-arm gain under joint person resampling, the form the registered reporting minimum names. For first attempt two of the four contrasts include zero (identity swap; pseudo-channel, whose point sits on the wrong side), so the cell is not separated from its controls. The two scalar control summaries are sensitivities: against the repetition medians (med.) the lower bound clears every control and the cell passes conditions 1–3 as historically implemented; against the event-pooled points (pool.) it clears none. The identity swap preserves item and duration-bin structure, so its retention points to item-level information; the pseudo-channel is one fixed global permutation and supports no mechanism claim. The same scalar dual read leaves all six primary cells unchanged (Table 3), where every primary bound already fails condition 1. For duration the interval lies inside the region under every reading (precise null); its paired contrasts are in the release. Repetition ranges (5 cut positions × 4 Monte Carlo seeds) are stability diagnostics only. Bottom: the M4 − M3 duration trace vanishes when the temporal block is rotated within person or swapped across persons; it is content-dependent, but it is a quarter of δ and its repetition interval includes zero. No within-person utility claim is licensed.

4.6. **What the block does predict instead.** Aimed at measurement-irrelevant context on the same folds and units, the facial block behaves differently (Table 5; Figure 6c). Its contrast for administration mode (whether the session was run in person in early 2020 or over Zoom in late 2020) is positive as a point median (+0.0154, 90% of repetitions positive) and larger than for any cognitive outcome, whereas task version is negative (−0.0034). This is the direction one would expect if the block partly encodes the recording situation (camera path, lighting, session context) rather than examinee cognition, and it sits naturally beside the unattributed person-stable signature reported by Lee and Wang (2026). Mode is constant within person, however, so the honest inferential unit is the person ($n = 169$, split 42/127), and at that level the event-weighted interval includes zero (+0.0146; interval −0.031 to +0.060), as do all paired mode-minus-cognition differences (+0.018 to +0.034). We therefore

FIGURE 6. Residual Signals: Within-Person Cells and the Context Benchmark



Note. (a) Same-person-future M3 − M2. Filled points and bars are repetition medians and stability ranges (the misalignment arm's upper range is truncated, arrow); the open circles and thin bars nudged below the observed rows are the registered event-weighted participant-cluster bootstrap of the AMD-5 addendum release, the object on which these cells are adjudicated; shaded bands are equivalence regions. (b) The M4 − M3 duration gain under temporal-block corruptions (repetition ranges). (c) The same contrast aimed at recording context versus cognition; the orange bar below the administration-mode row is the person-level bootstrap interval, which includes zero.

report the fingerprint reading as descriptive, and its evidential role here is comparative: whatever the context tendency ultimately is, nothing in the cognitive column outperformed even this unadjudicated context signal.

Table 5. The Same Representation Aimed at Cognition Versus Recording Context

Target type	Target	Mdn	Split range	Pr(>0)
Cognition	Next-item first attempt	$-.0036$	[-.0062, -.0024]	.00
Cognition	Next-item log duration	$-.0041$	[-.0062, -.0021]	.00
Cognition	Post-error help seeking	$-.0052$	[-.0123, .0024]	.08
Cognition	Post-error skipping	$-.0155$	[-.0201, -.0093]	.00
Context	Task version	$-.0034$	[-.0052, -.0024]	.00
Context	Administration mode (in person vs. Zoom)	.0154	[-.0101, .0266]	.90
Person-level mode interval (95%, $n = 169$): [-.031, .060]; includes zero.				
Paired mode-minus-cognition differences: .018 to .034; all include zero.				

Note. Same contrast (M3 − M2), same folds, same units. As a point median the facial block predicts which administration mode a session belongs to (in person, February–March 2020, versus over Zoom, September 2020–January 2021) better than it predicts any cognitive target, while adding nothing to cognition; this is the direction expected if the block partly functions as a recording-context fingerprint. Because mode is constant within person, the honest uncertainty level is the person ($n = 169$, 42/127 split), and at that level the interval includes zero, so the fingerprint reading is reported as descriptive rather than as an adjudicated claim.

4.7. **Robustness.** The conclusions are insensitive to every audited analytic choice. Restoring the ninth channel (Joy) moves the primary medians by at most 0.0012 bits and changes no verdict, and the null calibration run under the nine-channel scope mirrors the primary one (negative in 99 of 100 first-attempt replicates, with a negative 97.5th percentile). Judging under the pre-amendment equivalence regions changes no verdict. Replacing the event-weighted inferential functional with the equal-person mean of person means shifts the largest cell by 0.0034 bits (post-error skipping, -0.0148 versus -0.0114) and changes no verdict. The retrospective result is stable across observed-block sizes $k = 5$ to 18 (Appendix D). Calibration slopes and intercepts stayed within the gate's tolerances without rescuing any facial cell, and the secondary metrics (Brier score, RMSE) track the log-score conclusions throughout (Appendix D). The registered scenario record therefore reads *redundancy with a descriptive fingerprint*: no evaluated facial representation passed any gate in any primary cell; the within-person positives failed control separation or fell below the equivalence threshold; and the context tendency, while suggestive, is not resolvable at the person level at this sample size.

5. **Discussion**

We asked whether commercial facial-affect channels add out-of-sample predictive information to a cognitive assessment's own behavioral log, and we evaluated the question with a benchmark constructed so that a spurious gain would be unlikely to survive it. The verdicts were uniform: in every primary cell the facial summary made held-out prediction slightly worse, by amounts indistinguishable from the complexity penalty of adding the same number of noise columns; four of six cells are precise nulls within their equivalence regions and two are inconclusive at the boundary, identically under the pre-amendment and confirmed regions. Richer temporal and person-normalized facial representations performed worse still, and held-out predictive performance peaked at the last behavioral rung (M2) for every outcome. The few positive traces that exist are not separated from their own corrupted controls or fall inside their equivalence regions.

5.1. **Why the null is informative.** A null result carries weight only when the design could have detected the alternative and would have accepted it. Three features of the design support that reading here. First, resolution: the injection–recovery simulation showed that the pipeline reliably detects synthetic first-attempt effects of 0.0038 bits per event, an order of magnitude smaller than the behavioral yardsticks, so the binary nulls are bounded rather than merely nonsignificant (Lakens, 2017); the Gaussian outcomes are bounded by their much larger yardstick-based regions rather than by a calibrated recovery floor. Second, sensitivity to artifact: the same-item probe shows that the pipeline reports large gains (+0.255 bits) as soon as a leakage path is opened, so the machinery does not simply shrink every contrast toward zero. Third, a strong baseline: the behavioral log's own increments (+0.036 to +0.094 bits) demonstrate that this corpus contains extractable process information; the facial block simply does not add to it. Together these features support a statement stronger than the mere absence of an observed gain: to the resolution certified by the simulations, incremental information of this kind is absent from these data.

5.2. **The sensor-context reading.** Descriptively, the facial block predicted whether a session was run in person or over Zoom better than it predicted any cognitive outcome. A block that tracks the recording situation (camera path, lighting, compression, session context) more strongly than it tracks cognition is behaving like a sensor of that situation, and this converges with the finding of Lee and Wang (2026) that the same channels carry a stable within-session person signature whose source that design could not separate from morphology, expressivity, hardware, and calibration. Two reporting decisions should be noted. The mode tendency holds as a point-median description but fails the person-level bootstrap that the mode variable's structure demands ($n = 169$; the interval includes zero), so it enters the record as a description rather than an adjudicated finding; the downgrade was forced by our own review process, which we regard as the process operating as intended. Its evidential role is accordingly comparative: the cognitive contrasts could not match even a context signal that itself fails person-level adjudication. Under both readings, what the channels carry is at least as consistent with the recording context (a bundle of mode, calendar time, task mix, and capture path that this design cannot separate) as with anything about the examinee's cognitive process.

5.3. **Implications for practice.** For measurement practice the immediate implication is narrow and concrete: in this corpus, with one commercial pipeline's channel scores as delivered and with the representations that deployment arguments actually propose (summaries, temporal organization, person normalization), facial-affect channels did not improve predictive models of examinee behavior. The assessment log, which carries no added privacy cost, already contained everything these models could use. Programs considering emotion-sensing additions for proctored or tutoring contexts should therefore require the kind of evidence this benchmark formalizes before purchase: a frozen-in-advance incremental contrast against the full behavioral log, scored on held-out examinees, with corrupted-signal controls and a disclosed leakage audit (Hand, 2006; Kapoor & Narayanan, 2023). The positive control supplies a concrete reviewing criterion. Because a same-item alignment error inflates apparent value by a quarter bit per event on duration, seven times what the strongest single behavioral predictor contributes to that outcome, a vendor or study claiming facial gains should be asked to report their same-item probe and to show that the claimed gain exceeds their corrupted controls. On the evidence here, a gain reported without such safeguards is more plausibly attributed to artifact than to affect.

5.4. **A template for other process channels.** None of the benchmark's components is specific to facial data. Any candidate process channel (keystrokes, mouse dynamics, gaze, physiological streams) faces the same inferential hazards: strong behavioral baselines, high-volume weak-signal data, within-person regimes that invite overinterpretation, and alignment artifacts. The template transfers as a set of requirements: fix the estimand in bits on a strictly proper score; ladder the representations so that every contrast is paired; separate person-transport from within-person regimes explicitly; calibrate an equivalence region by injection–recovery before unblinding; register leakage metadata for every feature, with automated checks; run corrupted-channel controls and one deliberately leaky positive control in the same release; keep the gate's conditions free of any preferred direction; and submit the whole object to adversarial audits with the authority to force reruns. Our audits found real defects in the synthesis and reporting layer and in parts of the implementation, and the corrected record is stronger for each of them; we recommend the practice as a default for benchmark studies generally.

5.5. **Limitations.** Six limitations bound the claims. First, instrumentation: the results concern one commercial system's channel scores as delivered; a different pipeline, raw facial action units, or video representations learned end to end could in principle carry information these channels discard. Second, population and context: 169 examinees, one 25-item assessment, two administration modes collected in successive, non-overlapping pandemic-era windows; transport beyond this setting is exactly what the benchmark warns against assuming. Third, model class: ΔU is defined relative to ridge-penalized generalized linear models with inner-CV penalty selection; a registered gradient-boosting benchmark was planned but not executed, so what a nonlinear learner would extract, and whether it would separate from equally nonlinear controls, remains unexplored, and no conclusion here speaks to it. Fourth, resolution and one adjudication that postdates the freeze: the two first-attempt transport cells are inconclusive at the equivalence boundary, and the positive within-person cell passes the gate under the historical scalar control summary while the paired control contrasts, the form the registered reporting minimum names, resolve it as not separated. Every reading of that cell's condition 3 was computed after the interval existed, which is why all of them are released and reported rather than one being elected. The design certifies bounds rather than zeros, and $n = 169$ caps the person-level resolution of the mode comparison. Fifth, one deviation from our own freeze discipline stands on the record: the equivalence floor was confirmed after provisional gate results existed (AMD-2); we mitigated it by dual reporting, under which every verdict is identical. Sixth, outcomes are behavioral rather than affective: whether these channels predict self-reported emotion is a different question, untouched here, and nothing in our results identifies any subjective emotional state.

5.6. **Conclusion.** Measured against everything a cognitive assessment already logs, and judged by rules fixed so as not to favor any particular answer, no facial representation earned a usable predictive gain that survived its adversarial controls in the evaluated model classes and regimes, whether the prediction was for new examinees or for the same examinee later in the session. Four primary comparisons are bounded nulls, two are inconclusive at the boundary, and the one within-person gain that is bounded away from zero, the only comparison to pass the original screening rule, cannot be statistically separated from corrupted signals that carry no facial content. What the channels do carry remains unidentified by this design; a stable person signature and a descriptive recording-context tendency are the leading candidates, and nothing here adjudicates between them. We conclude that the burden of proof for emotion sensing in assessment rests on demonstrations that survive an adversarial evaluation of this kind, and this article provides the evaluation standard, the tooling, and one complete, corrected worked example of a candidate representation that does not survive it.

Software and reproducibility. All analyses used R 4.6.0 (R Core Team, 2026) with glmnet 5.0 as the fitting engine of every model (Friedman et al., 2010); figures were produced with ggplot2. A replication package rebuilds all twenty tables and six of the seven figures in this article and its supplement from the frozen analysis releases, and carries the annotated analysis code, the design contracts with their timestamps, the public fold architecture and custodian-only empirical-seed contract, the leakage register and its automated checks, a technical report developing the method, and ten operational verification gates, among them the manuscript's own number-verification script ported unchanged and rewired to the released aggregates. A separate meta-test demonstrates gate sensitivity with thirteen planted defects and an untouched-release control [REPOSITORY URL TO BE ADDED]. Figure 1 is not rebuilt: three of its four panels are at participant grain, and the omission is registered in the package rather than left to be discovered.

Data availability. No participant data are shared at any grain. The corpus was collected under an institutional approval that does not permit redistribution, and there is no route by which a third party can obtain it: the analysis pipeline reported here cannot be rerun outside the original team. What the replication package distributes is the analysis code together with the aggregate result releases, whose finest grain is a design cell rather than a person. The governing rule, applied file by file and enforced against the manifest-defined staged release, is that no released object may contain or permit reconstruction of a value at the grain of a participant or of a participant-by-item cell; the package records the decision for each object, including the ten it withholds. The upstream measurement releases of Lee and Wang (2026) are consumed through a hash-verified read-only interface and are governed by that study's terms.

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Online Supplement

Do Facial-Affect Process Data Improve Cognitive Measurement?
An Adversarial Out-of-Sample Benchmark

JoonHo Lee and Yurou Wang

This supplement accompanies the main text. Section, equation, table, and figure numbers below are prefixed by the appendix letter (A.1, B.1, ...); references to the main text use its unprefixed numbering. The author-side build verifier source-locks an enumerated set of numeric claims to the frozen releases; the public package independently re-checks 25 selected printed values under a typed, source-located contract.

Contents

Appendix A. Corpus and Measurement Detail

This appendix documents the corpus facts that the main text summarizes: the administration-mode composition, what is and is not known about the capture paths, and the coverage pattern. Counts derived from the frozen analysis releases are marked as such; the remaining descriptives are archival, recomputed at build time from the immutable raw stores by a script shipped with the manuscript (`figures/src/00_extract_cache.R`) and compared against the prose by the number-verification gate (Appendix G describes what that gate can and cannot check).

A.1. **Administration modes.** Data collection began in person on February 18, 2020, and was halted by the COVID-19 campus closure; the last in-person session ran on March 13, 2020. Collection resumed remotely on September 2, 2020, with sessions administered over Zoom, and ended on January 29, 2021. The two windows do not overlap and no participant appears in both, so the session timestamps partition the cohort exactly, and administration mode and recording wave are the same partition of these data. The evidential basis for the mode labels is twofold and not uniform: 162 of the 169 analytic participants have a survey record whose administration-setting field agrees with the timestamp partition (42 on site, 120 online); the remaining seven, all in the late window, lack a survey record and are classified as remote from their session timestamps alone. Task version is taken from the historical person table, so for those seven participants it, too, is historically rather than survey-confirmed. Both source publications describe the collection as conducted online over Zoom (Wang et al., 2021, 2025); the in-person window appears only in the archival record, and this article reports the archival account. Table A.1 gives the composition.

Table A.1. Composition of the analytic cohort by administration mode

Quantity	In person	Over Zoom	Total
Participants	42	127	169
verbal version	7	55	62
mathematics version	35	72	107
Observed transitions	952	3,047	3,999
Unobserved transitions	56	1	57

Note. In-person sessions ran from February 18 to March 13, 2020, and were halted by the COVID-19 campus closure; remote sessions ran over Zoom from September 2, 2020, to January 29, 2021. The two waves partition the cohort exactly (no participant appears in both), so administration mode and recording wave are the same variable in these data. Of the 169 participants, 162 have a survey record whose setting field agrees with this partition; seven late-window participants lack a survey record and are classified as remote from their session timestamps, with task version taken from the historical person table. These counts are archival descriptives recomputed at build time from the immutable stores; they are not part of the frozen analysis releases.

Three further facts about the in-person subset deserve explicit statement. First, the capture path of the in-person sessions is not documented in the archival record; the remote sessions read the participant's video window inside Zoom on the study computer. The inference software was identical in both modes and its internal tick-clock cadence was the same (a median of 160 ms between samples, about six per second; the realized within-item wall-clock density is close to 5.8 Hz in both modes, the figure reported by Lee and Wang (2026), and the two numbers are different estimands of the same stream, not competing estimates). The optical and compression paths cannot be assumed comparable, and no finer hardware metadata were recorded. Second, the in-person recordings were more often truncated near the end of the session: 56 of the 57 unobserved transitions belong to in-person participants, concentrated at the last item positions. Third, mode is partially confounded with task version (7 of 42 in-person participants took the verbal version, against 55 of 127 remote participants) and fully confounded with calendar time. These confounds are why mode never enters any model as a predictor, why the mode-transport regime is reported per direction with the small in-person wave flagged as the low-information direction, and why the mode target of the context benchmark (main text, Section 4.6) is adjudicated at the person level.

A.2. **Channels and their sparsity.** Eight of the nine vendor source streams take integer values from 0 to 100; the ninth, *Valence*, is signed, with an observed range of roughly -100 to 100. The historical crosswalk of Lee and Wang (2026) maps vendor *Surprise* to the project-derived analysis label *Anxious*, and vendor *Engagement* to *Attentive* through the historical intermediate name *Danger*. It maps the absolute magnitude of vendor *Valence* to the project-derived label *Negative*; valence direction is therefore not retained in that analysis stream. These are project analysis labels, not vendor product names. Activation is accordingly defined throughout as a nonzero absolute value. Most channels are silent on most samples and activation arrives in bursts (Figure 1 of the main text shows an illustrative session). Across the corpus's aligned samples, *Attentive* is active on roughly half of all samples, *Anxious* on just under 40%, *Negative* on roughly 16%, and *Joy* on well under one percent. This sparsity is one reason the facial summary block records occurrence and active share explicitly rather than relying on means alone, and the near-silence of *Joy* is consistent with its exclusion from the primary block on the person-ordering evidence of Lee and Wang (2026).

A.3. **Exemplar.** The exemplar participant of main-text Figure 1 was selected for legibility, not typicality of any focal result: an illustrative remote-mode, mathematics-version session with all 25 items observed and a mix of correct first attempts, retries, hints, and one skip. No identifier is shown; the cached extract used to draw the figure carries only the label `exemplar`.

Appendix B. Design Contracts

This appendix restates, in prose, the machine-readable contracts in the `config/` directory of the replication package, frozen before any facial result was computed. The configuration files themselves, with timestamps, are authoritative.

B.1. **Estimand.** For a cell (outcome $\times$ regime $\times$ repetition), each rung's out-of-fold predictions are scored with the base-2 logarithmic score: Bernoulli log-loss for binary outcomes, and for log duration the Gaussian density score $\log_2 \phi\{(y-\hat{\mu})/\hat{\sigma}_f\} - \log_2 \hat{\sigma}_f$, where $\hat{\sigma}_f$ is the training-residual standard deviation of the row's own fold f. The estimand is the paired difference in mean score between adjacent rungs (main text, Equation (1)), in usable bits per event. ΔU is model-class relative: it measures what a ridge-penalized GLM analyst can extract under honest folds, not the mutual information the channel "contains." Positive ΔU means that the added block contributed predictive information net of its complexity penalty on held-out data.

B.2. **Outcome contract.** Prospective outcomes at item $i+1$: first-attempt accuracy (`correct1`); log duration; post-error help seeking ($N_{\text{hint}} > 0$ and no skip, defined only when the first attempt failed); post-error skipping (skip after a failed first attempt; skip takes precedence, making the post-error categories exclusive). Retrospective outcomes: first-attempt accuracy and log duration of each remaining item position, given observed blocks of $k \in \{5, 8, 12, 15, 18\}$ positions ($k = 12$ primary). All definitions were verified cell-by-cell against the source release before freezing.

B.3. **Ladder, channel scope, and M4 eligibility.** The six rungs and their counts appear in main-text Table 2. The implemented facial magnitude summary is the source study's strict-positive magnitude: the conditional mean of $\log|\text{value}|$ over samples with $|\text{value}| > 1$, zero when no sample is strictly active. The contract's original wording ("maximum activation magnitude") misdescribed this implemented feature and is corrected under AMD-5; no model or release is refit by the correction. The facial block covers the eight channels with `m3_primary = TRUE` in the channel-eligibility contract (Joy is sensitivity-only, following the channel evidence of Lee and Wang, 2026); enforcement is programmatic (`pb_rung_columns(exclude_channels)`), and the nine-channel representation survives only as the labeled arm `observed_sensitivity_9ch` (Appendix E). M4 candidates (per-channel lag-1 autocorrelation, effective-sample-size ratio, variability flag) entered only if an outcome-free eligibility test passed: the feature's within-person deviation profile across items had to agree between random participant halves more strongly than an item-rotation surrogate, a test that never touches any outcome; of the 27 candidates, 25 passed, including all 24 primary-channel (non-Joy) candidates. M5 person means and deviations are computed inside each training fold by `pb_person_norm_transform()`; held-out persons receive the training grand mean.

B.4. **Fold architecture.** Person held out: 5 folds by person, 50 repetitions (the public `config/folds/_SEEDS.json` records architecture and counts; the empirical seed and assignments remain custodian-only). Public non-empirical phase seed bases are recorded in the corresponding release `_SUCCESS.json`. Item-block held out: 5 contiguous position blocks. Crossed: test cells share neither person nor item with any training row; rows sharing either are excluded. Same-person-future: train on positions $1..c$, test on $c{+}1..25$, $c \in \{8, 10, 12, 14, 16\}$ crossed with four Monte Carlo seeds, giving 20 repetitions (5 cut positions $\times$ 4 seeds). Mode and task transport: single directional splits, reported per direction (the release files name the mode regime `wave_held_out`). All rungs within a cell share rows, folds, and seeds.

B.5. **Adjudication gate.** Condition 1 (as amended by AMD-4/B-05): the inferential interval (the event-weighted participant-cluster bootstrap) lies outside the equivalence region $[-\delta, +\delta]$ (Appendix C); the repeated-split range is released as a stability diagnostic and never adjudicates equivalence. Phase 12 had nonetheless judged the two within-person cells on repetition quantiles; the second external review flagged the deviation, and the registered bootstrap for those cells is supplied by the AMD-5 addendum release, which also refits the corrupted arms under the identical seeds and releases the paired control contrasts that adjudicate condition 3. Condition 2: calibration does not deteriorate; the added rung's calibration slope must be no farther from 1 than the base rung's (tolerance .05) and its intercept no farther from 0 (tolerance .10), defined for binary outcomes. Condition 3: the observed gain is separated from the corrupted controls run in the same regime; a missing battery yields the named verdict `CONTROLS_NOT_EVALUATED`. The adjudicative form is the one the registered reporting minimum names, the paired adversarial control contrast: observed minus corrupted-arm gain under joint person resampling, met only if every contrast's lower bound exceeds zero. Two scalar summaries of the control arms are released as sensitivities (repetition median, as the original implementation computed it, and event-pooled point, the functional of the observed bound); the scalar summaries agree everywhere except the same-person-future first-attempt cell (Appendix E). All three required, in order; named failure verdicts: precise null (inside region), inconclusive (straddles), negative-outside, ranking-only (calibration fail), artifact (controls not exceeded), controls-not-evaluated. Gate C (transport licensing) additionally requires at least five repetitions behind a positive interval and survival of the same battery in that regime; single-split point estimates license nothing.

B.6. **Amendment register.** **AMD-1** (pre-facial): the equivalence region was made outcome-specific ($\delta = 0.5\times$ the largest behavioral yardstick layer for that outcome) because a single global δ was indefensible across outcomes whose yardsticks span two orders of magnitude. **AMD-2** (post-provisional-gate; disclosed deviation): δ was floored at the design resolution A_3 (0.0036 bits as then calibrated on the 27-column block; recalibrated to 0.0038 on the 24-column primary block under AMD-4), the smallest injected effect recovered in $\geq 90\%$ of simulation replicates, raising the two first-attempt outcomes. Because this followed a provisional look at gate results, verdicts under both regions are released (Table C.3; main-text Table 3), and the deviation is listed as a limitation. **AMD-3** (first review response): channel scope enforced in code; fold-internal person normalization; duration-tertile identity swap; register wording aligned to implementations; single-repetition licensing closed; calibration intercept added to the gate. **AMD-4** (second, external review response): the Phase-6 null and injection–recovery calibration aligned to the 24-column primary block with the nine-channel scope retained as a paired sensitivity (A_3 recomputed); the participant-cluster bootstrap made event-weighted so its interval estimates the registered bits-per-event functional (equal-person means kept only as a labeled sensitivity); gate condition 1 moved onto that inferential interval, with split ranges demoted to stability diagnostics; matched retrospective control batteries added and a missing battery routed to the named verdict `CONTROLS_NOT_EVALUATED`; the retrospective predictors registered (`safe_observed_block`) with a runtime column assertion; the identity swap made a guaranteed derangement; the same-person gate given its calibration intercept; and the clean-run driver made fail-closed with relocatable roots. **AMD-5** (second external review response and its focused re-review, this build): the same-person-future M3 − M2 cells received their registered inferential object in the addendum release `spf-person-bootstrap_v1`, whose refits (observed and all four corrupted arms, both outcomes) reproduce every released repetition to 10^{-9} before release. Condition 3 for these cells is adjudicated on the released paired contrasts (first attempt: the identity-swap and pseudo-channel contrasts include zero, verdict artifact-not-separated; duration: precise null), with the two scalar control summaries released as sensitivities; the first-attempt cell passes conditions 1–3 under the historical repetition-median rule, and that split is disclosed wherever the cell is reported. The M3 magnitude wording was corrected to the implemented conditional mean log magnitude, on the re-review also in the frozen ladder contract itself and the feature builder's header;

and the archival *Negative* channel description was corrected to the signed, sign-folded convention. No frozen release is modified and no released number changes.

B.7. **Register of analyses deliberately not run.** (1) A gradient-boosting upper benchmark, registered as permissible when the penalized GLM converges cleanly, was planned but not executed; no claim is made about what a nonlinear learner would extract, and the main text's model-class limitation says so. (An earlier stronger waiver rationale was withdrawn under external review.) (2) The nested-versus-non-nested CV optimism comparison was not run: penalty selection is nested by construction, and selection optimism inflates gains, which is the conservative direction for an all-negative result. (3) The `coverage_only` control was struck as degenerate: recording-coverage terms appear in M0 of both rungs of every contrast, so a coverage-only arm reduces to M2 versus M2 by construction. All three entries carry their reasoning in `config/amendments.yml`.

Appendix C. Simulation Calibration

Both calibrations run on the real feature matrix with only the facial block manipulated and, following external review (AMD-4/B-01), on exactly the 24-column eight-channel block the primary analysis uses, with the 27-column nine-channel scope retained as a paired sensitivity. The original calibration was blind (run before any observed facial contrast existed); the scope-aligned rerun reuses the same design after the review found the original had calibrated the wrong block width.

C.1. **Null false-gain distribution.** The facial block was replaced by a person-shuffled copy within item (so item-level structure survives but person–outcome correspondence is destroyed) and the full pipeline was run 100 times per outcome family. Table C.1 summarizes: negative in 99 of 100 first-attempt replicates, with a negative 97.5th percentile in every scope. This is the complexity penalty of 24 uninformative columns under ridge with inner-CV penalty selection, and it is the correct reference point for the observed contrasts: an added block that carries no information does not center at zero; it centers near -0.003 bits. The observed primary contrasts (main-text Table 3) are statistically indistinguishable from this distribution.

Table C.1. Null false-gain distribution: M3 − M2 when the facial block is person-shuffled within item, by calibration scope

Outcome	Scope	n reps	Mean	$q_{.975}$	Max
first_attempt_to	primary_8ch	100	−.0031	−.0002	.0002
log_duration_to	primary_8ch	100	−.0045	−.0014	−.0006
first_attempt_to	sensitivity_9ch	100	−.0034	−.0003	.0001
log_duration_to	sensitivity_9ch	100	−.0051	−.0021	−.0014

Note. The primary scope is the 24-column eight-channel block (AMD-4/B-01); the 27-column nine-channel scope is a paired sensitivity. The 97.5th percentile is negative in every scope; a single first-attempt replicate crosses zero at +0.0002.

C.2. **Injection–recovery and the design resolution.** Synthetic effects of known size were injected by coupling one facial carrier column to the outcome at strengths β and re-running the pipeline. Table C.2 reports the recovered ΔU and the proportion of replicates with positive recovery. $A_3 = 0.0038$ bits per event is the smallest injected effect recovered in at least 90% of replicates and serves as the design-resolution anchor: verdicts about contrasts smaller than A_3 are claims the design cannot reliably make, which is the rationale for AMD-2's floor. A_3 is a binary prospective resolution anchor (the injection design uses the first-attempt family), and it only ever floors the binary outcomes; the Gaussian outcomes' regions are set by their far larger behavioral yardsticks.

Table C.2. Injection–recovery curve: detected ΔU by injected effect strength

Injected β	Median bits	$q_{.025}$	Pr(> 0)
.00	.0001	−.0009	.55
.10	.0003	−.0006	.65
.20	.0038	.0009	1.00
.35	.0171	.0116	1.00
.50	.0368	.0272	1.00

Note. A_3 is the smallest injected effect recovered in at least 90% of replicates.

C.3. **Equivalence regions.** Table C.3 lists both generations of δ per outcome. The AMD-2 raise affects the two first-attempt outcomes only; main-text Table 3 reports verdicts under both, and the pass count (0 of 6) is identical.

Table C.3. Equivalence half-widths per outcome under AMD-1 (outcome-specific yardstick) and AMD-2 (raised to the design resolution A_3)

Outcome	δ (AMD-1)	δ (AMD-2)	Raised	A_3
heldout_first_attempt	.0014	.0038	yes	.0038
heldout_log_duration	.0493	.0493	no	.0038
next_first_attempt	.0008	.0038	yes	.0038
next_log_duration	.0181	.0181	no	.0038
next_posterror_help	.0468	.0468	no	.0038
next_posterror_skip	.0423	.0423	no	.0038

Note. AMD-2 was confirmed after provisional gate results existed; the main text reports verdicts under both regions.

Appendix D. Complete Benchmark Results

D.1. **Primary regimes, all arms.** Table D.1 reports every arm run in the two primary regimes (person held out; retrospective block $k = 12$): the observed facial block, the four corrupted-facial controls, the matched-observed companions for row-dropping controls, the mean-only/deviation-only ablation, the same-item leakage probe, and the nine-channel sensitivity arm. The same-item probe is the only arm with a positive interval anywhere, and it is leaky by construction.

Table D.1. Complete M3 − M2 summary, primary regimes, all arms

Outcome	Arm	Reps	Mdn	$q_{.025}$	$q_{.975}$	Pr(> 0)
n:first.att	circ.shift	20	−.0059	−.0079	−.0034	.00
n:first.att	dev.only	20	−.0057	−.0075	−.0034	.00
n:first.att	id.swap	20	−.0011	−.0026	.0001	.05
n:first.att	m.obs:circ.shift	20	−.0039	−.0061	−.0024	.00
n:first.att	m.obs:id.swap	20	−.0039	−.0061	−.0024	.00
n:first.att	m.obs:misalign +1	20	−.0039	−.0058	−.0026	.00
n:first.att	m.obs:pseudo	20	−.0039	−.0061	−.0024	.00
n:first.att	m.obs:same-item probe	20	−.0038	−.0058	−.0024	.00
n:first.att	mean only	20	−.0018	−.0024	−.0009	.00
n:first.att	misalign +1	20	−.0039	−.0059	−.0024	.00
n:first.att	observed	50	−.0036	−.0062	−.0024	.00
n:first.att	observed (9ch)	20	−.0043	−.0063	−.0025	.00
n:first.att	pseudo	20	−.0074	−.0100	−.0056	.00
n:first.att	same-item probe	20	.0246	.0205	.0274	1.00
n:log.dur	circ.shift	20	−.0044	−.0078	−.0020	.00
n:log.dur	dev.only	20	−.0100	−.0123	−.0086	.00

Outcome	Arm	Reps	Mdn	$q_{.025}$	$q_{.975}$	Pr(> 0)
n:log.dur	id.swap	20	−.0053	−.0076	−.0034	.00
n:log.dur	m.obs:circ.shift	20	−.0037	−.0053	−.0023	.00
n:log.dur	m.obs:id.swap	20	−.0037	−.0053	−.0023	.00
n:log.dur	m.obs:misalign +1	20	−.0044	−.0057	−.0022	.00
n:log.dur	m.obs:pseudo	20	−.0037	−.0053	−.0023	.00
n:log.dur	m.obs:same-item probe	20	−.0035	−.0051	−.0018	.00
n:log.dur	mean only	20	−.0106	−.0114	−.0082	.00
n:log.dur	misalign +1	20	−.0067	−.0090	−.0039	.00
n:log.dur	observed	50	−.0041	−.0062	−.0021	.00
n:log.dur	observed (9ch)	20	−.0049	−.0066	−.0027	.00
n:log.dur	pseudo	20	−.0057	−.0071	−.0037	.00
n:log.dur	same-item probe	20	.2554	.2491	.2630	1.00
n:pe.help	circ.shift	20	−.0097	−.0156	−.0040	.00
n:pe.help	dev.only	20	−.0041	−.0085	.0020	.20
n:pe.help	id.swap	20	−.0120	−.0176	−.0072	.00
n:pe.help	m.obs:circ.shift	20	−.0035	−.0081	.0030	.15
n:pe.help	m.obs:id.swap	20	−.0035	−.0081	.0030	.15
n:pe.help	m.obs:misalign +1	20	−.0022	−.0083	.0048	.25
n:pe.help	m.obs:pseudo	20	−.0035	−.0081	.0030	.15
n:pe.help	m.obs:same-item probe	20	−.0032	−.0087	.0030	.10
n:pe.help	mean only	20	−.0040	−.0056	−.0024	.00
n:pe.help	misalign +1	20	−.0085	−.0161	−.0024	.00
n:pe.help	observed	50	−.0052	−.0123	.0024	.08
n:pe.help	observed (9ch)	20	−.0064	−.0119	.0020	.15
n:pe.help	pseudo	20	−.0102	−.0141	−.0053	.00
n:pe.help	same-item probe	20	−.0041	−.0106	.0051	.20
n:pe.skip	circ.shift	20	−.0051	−.0104	.0023	.10
n:pe.skip	dev.only	20	−.0149	−.0209	−.0090	.00
n:pe.skip	id.swap	20	−.0108	−.0175	−.0073	.00
n:pe.skip	m.obs:circ.shift	20	−.0155	−.0177	−.0089	.00
n:pe.skip	m.obs:id.swap	20	−.0155	−.0177	−.0089	.00
n:pe.skip	m.obs:misalign +1	20	−.0109	−.0177	−.0065	.00
n:pe.skip	m.obs:pseudo	20	−.0155	−.0177	−.0089	.00
n:pe.skip	m.obs:same-item probe	20	−.0163	−.0189	−.0092	.00
n:pe.skip	mean only	20	−.0019	−.0052	−.0004	.00
n:pe.skip	misalign +1	20	−.0140	−.0202	−.0057	.00
n:pe.skip	observed	50	−.0155	−.0201	−.0093	.00
n:pe.skip	observed (9ch)	20	−.0166	−.0207	−.0098	.00
n:pe.skip	pseudo	20	−.0141	−.0208	−.0106	.00
n:pe.skip	same-item probe	20	−.0108	−.0201	−.0073	.00
h:first.att	circ.shift	20	−.0048	−.0079	−.0017	.00
h:first.att	id.swap	20	−.0094	−.0116	−.0069	.00
h:first.att	m.obs:misalign +1	20	−.0018	−.0051	.0019	.20

Outcome	Arm	Reps	Mdn	$q_{.025}$	$q_{.975}$	Pr(> 0)
h:first.att	m.obs:same-item probe	20	−.0022	−.0056	.0012	.10
h:first.att	misalign +1	20	−.0025	−.0069	.0000	.05
h:first.att	observed	50	−.0021	−.0066	.0007	.12
h:first.att	pseudo	20	−.0106	−.0149	−.0081	.00
h:first.att	same-item probe	20	−.0015	−.0047	.0020	.30
h:log.dur	circ.shift	20	−.0178	−.0335	−.0095	.00
h:log.dur	id.swap	20	−.0141	−.0241	−.0061	.00
h:log.dur	m.obs:misalign +1	20	−.0164	−.0260	−.0118	.00
h:log.dur	m.obs:same-item probe	20	−.0183	−.0263	−.0142	.00
h:log.dur	misalign +1	20	−.0204	−.0276	−.0153	.00
h:log.dur	observed	50	−.0184	−.0279	−.0100	.00
h:log.dur	pseudo	20	−.0312	−.0391	−.0184	.00
h:log.dur	same-item probe	20	−.0190	−.0262	−.0137	.00

Note. Labels: n:/h: = next-item / held-out outcome; m.obs: = matched-observed companion arm; full arm names as in the release files.

D.2. **Transport map.** Table D.2 gives the observed contrast in every regime, including the exploratory task-transport and single-direction mode splits. Under the licensing rules of Appendix B, no regime licenses a transport claim: the only positive multi-repetition intervals (same-person-future) fail their adjudication (Appendix E), and single-split regimes cannot license by construction.

TABLE D.2. Transport map: observed M3 − M2 in every regime

Regime	Outcome	Reps	Mdn	$q_{.025}$	$q_{.975}$
crossed	n:first.att	20	.0001	−.0083	.0061
item_block_held_out	n:first.att	1	−.0001	−.0001	−.0001
person_held_out	n:first.att	50	−.0036	−.0062	−.0024
same_person_future	n:first.att	20	.0114	.0012	.0804
task_test_literacy	n:first.att	1	−.0090	−.0090	−.0090
task_test_math	n:first.att	1	−.0015	−.0015	−.0015
wave_test_wave1	n:first.att	1	−.0026	−.0026	−.0026
wave_test_wave2	n:first.att	1	−.0129	−.0129	−.0129
crossed	n:log.dur	20	−.0017	−.0084	.0093
item_block_held_out	n:log.dur	1	.0008	.0008	.0008
person_held_out	n:log.dur	50	−.0041	−.0062	−.0021
same_person_future	n:log.dur	20	.0062	.0007	.0226
task_test_literacy	n:log.dur	1	−.0235	−.0235	−.0235
task_test_math	n:log.dur	1	−.0142	−.0142	−.0142

Regime	Outcome	Reps	Mdn	$q_{.025}$	$q_{.975}$
wave_test_wave1	n:log.dur	1	−.0056	−.0056	−.0056
wave_test_wave2	n:log.dur	1	−.0798	−.0798	−.0798
crossed	n:pe.help	20	.0005	−.0207	.0153
item_block_held_out	n:pe.help	1	.0047	.0047	.0047
person_held_out	n:pe.help	50	−.0052	−.0123	.0024
same_person_future	n:pe.help	20	−.0073	−.0127	.0019
task_test_literacy	n:pe.help	1	.0071	.0071	.0071
task_test_math	n:pe.help	1	−.0089	−.0089	−.0089
wave_test_wave1	n:pe.help	1	.0023	.0023	.0023
wave_test_wave2	n:pe.help	1	−.0175	−.0175	−.0175
crossed	n:pe.skip	20	−.0184	−.0308	−.0103
item_block_held_out	n:pe.skip	1	−.0137	−.0137	−.0137
person_held_out	n:pe.skip	50	−.0155	−.0201	−.0093
same_person_future	n:pe.skip	20	−.0023	−.0098	.0309
task_test_literacy	n:pe.skip	1	−.0106	−.0106	−.0106
task_test_math	n:pe.skip	1	.0073	.0073	.0073
wave_test_wave1	n:pe.skip	1	−.0229	−.0229	−.0229
wave_test_wave2	n:pe.skip	1	−.0396	−.0396	−.0396
retro_block_05	h:first.att	20	−.0020	−.0043	.0008
retro_block_08	h:first.att	20	−.0041	−.0059	−.0018
retro_block_12	h:first.att	50	−.0021	−.0066	.0007
retro_block_15	h:first.att	20	−.0035	−.0062	.0006
retro_block_18	h:first.att	20	−.0012	−.0051	.0026
retro_block_05	h:log.dur	20	−.0246	−.0328	−.0143
retro_block_08	h:log.dur	20	−.0185	−.0273	−.0098
retro_block_12	h:log.dur	50	−.0184	−.0279	−.0100
retro_block_15	h:log.dur	20	−.0115	−.0202	−.0056
retro_block_18	h:log.dur	20	−.0068	−.0141	.0006

Note. Row labels name the test set: `wave_test_wave1` tests on the in-person wave (training on Zoom) and `wave_test_wave2` the reverse; `task_test_literacy` tests on the verbal (literacy) version. Administration mode and wave are the same partition in these data. The same-person-future rows are the Phase-8 benchmark estimates; the Phase-12 battery of Appendix E independently refits the same folds under different registered tuning seeds, which is why its summaries differ slightly (matched-row discipline applies to within-battery control comparisons, not to this release-to-release difference).

D.3. **Temporal and personalized rungs; frontier.** Table D.3 reports all M4 − M3 and M5 − M4 cells from the Phase-9 benchmark release; for the same-person-future duration cell, Appendix E re-estimates the contrast with the corrected temporal battery on matched rows, and the two estimates are both printed. Table D.4 gives the codelength frontier (rungs M0 through M4) behind main-text Figure 5. M5 values were produced by the corrected, fold-internal person normalization (Appendix F, finding F-03).

Table D.3. M4 and M5 contrasts, all cells

Regime	Contrast	Outcome	Reps	Mdn	$q_{.025}$	$q_{.975}$	Pr(> 0)
person_held_out	M4_minus_M3	n:first.att	50	−.0017	−.0028	−.0009	.00
same_person_future	M4_minus_M3	n:first.att	20	.0004	−.0017	.0204	.55
person_held_out	M4_minus_M3	n:log.dur	50	−.0016	−.0028	−.0002	.02
same_person_future	M4_minus_M3	n:log.dur	20	.0042	.0010	.0244	1.00
person_held_out	M4_minus_M3	n:pe.help	50	−.0053	−.0111	−.0015	.00
same_person_future	M4_minus_M3	n:pe.help	20	−.0090	−.0100	−.0044	.00
person_held_out	M4_minus_M3	n:pe.skip	50	−.0045	−.0087	−.0019	.00
same_person_future	M4_minus_M3	n:pe.skip	20	−.0048	−.0076	.0243	.35
person_held_out	M5_minus_M4	n:first.att	20	−.0024	−.0036	−.0011	.00
person_held_out	M5_minus_M4	n:log.dur	20	−.0184	−.0203	−.0156	.00
person_held_out	M5_minus_M4	n:pe.help	20	−.0084	−.0116	−.0012	.00
person_held_out	M5_minus_M4	n:pe.skip	20	−.0055	−.0101	−.0017	.00

Note. Phase-9 benchmark estimates. For the same-person-future duration cell, Appendix E reports the corrected temporal-battery estimate (Table E.2), whose repetition interval includes zero; the two releases refit identical folds under different registered tuning seeds, which is why their summaries differ slightly.

TABLE D.4. Held-out codelength frontier (rungs M0–M4), full table

Outcome	Rung	k	Median codelength	SE
next_first_attempt	M0	6	.9606	.0003
next_first_attempt	M1	8	.9591	.0003
next_first_attempt	M2	15	.9573	.0003
next_first_attempt	M3	39	.9615	.0004
next_first_attempt	M4	63	.9636	.0005
next_log_duration	M0	6	1.6366	.0006
next_log_duration	M1	8	1.5998	.0005
next_log_duration	M2	15	1.5944	.0007
next_log_duration	M3	39	1.5991	.0006
next_log_duration	M4	63	1.5999	.0006
next_posterror_help	M0	6	.8999	.0011
next_posterror_help	M1	8	.8959	.0011
next_posterror_help	M2	15	.8029	.0007
next_posterror_help	M3	39	.8067	.0014
next_posterror_help	M4	63	.8114	.0015
next_posterror_skip	M0	6	.6585	.0013
next_posterror_skip	M1	8	.6590	.0012
next_posterror_skip	M2	15	.5760	.0009
next_posterror_skip	M3	39	.5896	.0008
next_posterror_skip	M4	63	.5930	.0010

Note. Minimum per outcome is at M2 in every case (circled in the main-text figure).

D.4. **Secondary metrics.** Table D.5 reports the reporting-minimum metrics per rung for the primary cells: Brier score (binary), RMSE (continuous), and calibration intercept and slope. No secondary metric reverses any verdict; in the two post-error cells some individual calibration numbers sit nominally closer to their ideals under M3 than under M2, but the log-score contrast, the gate object, remains negative there, and the calibration condition of the gate is a tolerance test rather than a comparison of these medians.

TABLE D.5. Secondary metrics per rung (medians across repetitions), primary cells, observed arm

Outcome	Rung	$\log_2$ score	Brier	RMSE	Cal. int.	Cal. slope
n:first.att	M2	−.9574	.2354	—	−.0058	1.0136
n:first.att	M3	−.9613	.2367	—	.0077	.9694
n:log.dur	M2	−1.5950	—	.7302	—	—
n:log.dur	M3	−1.5987	—	.7322	—	—
n:pe.help	M2	−.8022	.1894	—	.0453	1.0857

Outcome	Rung	$\log_2$ score	Brier	RMSE	Cal. int.	Cal. slope
n:pe.help	M3	$-.8070$	.1910	—	$-.0140$	.9757
n:pe.skip	M2	$-.5754$	.1230	—	.0737	1.0462
n:pe.skip	M3	$-.5909$	.1266	—	$-.0467$	.9533
h:first.att	M2	$-.9262$	.2253	—	$-.0036$	1.0079
h:first.att	M3	$-.9286$	.2260	—	$-.0169$	1.0281
h:log.dur	M2	-1.3575	—	.6189	—	—
h:log.dur	M3	-1.3773	—	.6262	—	—

Note. Metric medians are taken per metric across repetitions and can differ in the third decimal from the codelength medians of the frontier table, which are aggregated separately.

D.5. **Retrospective block-size curve.** Figure D.1 traces the retrospective contrast across observed-block sizes $k = 5$ to 18. All medians are negative at every k; the primary $k = 12$ is representative, not selected.

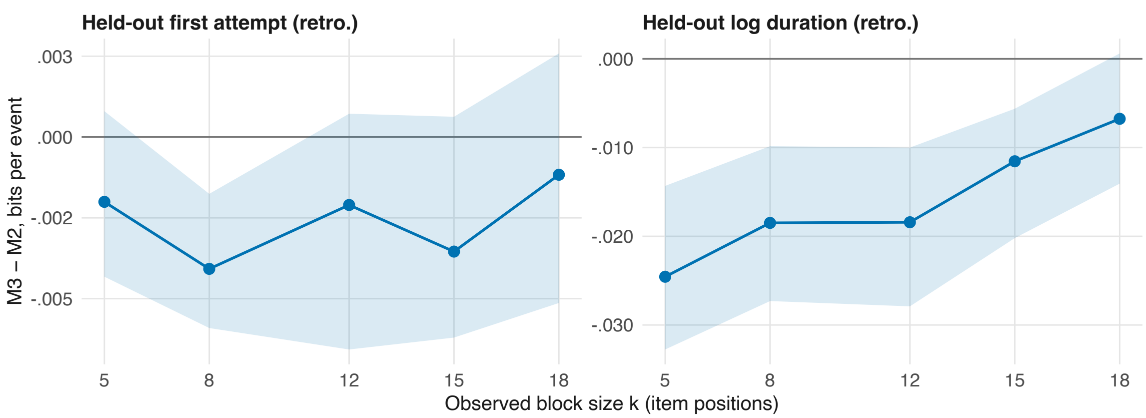

FIGURE D.1. Retrospective M3 − M2 by observed-block size k. Ribbons are 2.5–97.5% split ranges.

Appendix E. Adjudication of Residual Signals

This appendix reports the adjudication of the residual signals: the two within-person positive cells that the first audit surfaced, the temporal-block battery that replaced an unsupported artifact verdict, the person-level administration-mode bootstrap, and the channel-scope sensitivity. Two releases are involved. The Phase-12 release (`adjudication_v1`) carries the batteries and repetition summaries; the AMD-5 addendum (`spf-person-bootstrap_v1`), produced after the second external review found that Phase 12 had adjudicated the within-person cells on repetition quantiles, carries the registered event-weighted participant-cluster intervals for those cells and the gate re-read on the matching event-pooled control comparison.

E.1. **Same-person-future M3 − M2, full battery.** Table E.1 gives every arm for both outcomes; the repetition summaries it reports are stability diagnostics. The registered adjudication reads as follows. For first attempt, the event-weighted participant-cluster interval is $+0.0262$ $[+0.0197, +0.0333]$, outside the ± 0.0038 region, with calibration intact, so conditions 1 and 2 are met. Condition 3 is adjudicated on the paired adversarial contrasts (`spf-paired-contrasts.csv`): each corrupted arm was refit under the identical Phase-12 seeds, fidelity-gated the same way, and the observed-minus-control gain was bootstrapped under joint person resampling (2,000 resamples; the all-contrasts decision is replicated at 20,000 resamples before release). Two contrasts exclude zero (circular shift $+0.0066$ $[+0.0017, +0.0118]$; misalignment $+0.0289$ $[+0.0221, +0.0356]$) and two include it (identity swap $+0.0049$ $[-0.0003, +0.0099]$; pseudo-channel -0.0023 $[-0.0068, +0.0026]$), so the cell is *artifact-not-separated* (`ARTIFACT_CONTROLS_NOT_SEPARATED` in `spf-gate-revisited.csv`).

The two scalar control summaries are released as sensitivities, and this is the only cell in the study where their choice changes a reading. Against the repetition medians (largest $+0.0052$) the lower bound clears every control and the cell passes conditions 1–3 as historically implemented (`GATE_B_PASS_CONDITIONS_1_3`). Against the event-pooled points, the functional of the observed bound, it clears none (pseudo-channel $+0.0285$, identity swap $+0.0213$). Every one of these adjudications postdates the frozen release; the paired contrast is the form the registered reporting minimum names, and it is the reading the verdict rests on. On mechanism the design identifies less than the size pattern suggests. The identity swap preserves item identity and duration bin by construction (it retains every primary column's item mean exactly), so its near-full retention points to item-level information available to any block indexed to the same items. The pseudo-channel arm is a single fixed per-column permutation of the source cells: approximately marginal-matched on the evaluated transitions, preserving neither item structure nor within-row coherence, so the source of its matching gain is not identified (a fixed permutation can carry accidental correlations and can interact with tuning and shrinkage). Its evidential role is the negative one: gains of this size arise from blocks with no usable facial content. Applying the scalar dual read to the six primary cells changes none of their verdicts, since every primary bound already fails condition 1 (`primary-gate-both-comparators.csv`); paired contrasts are not computed there because no condition-3 reading can alter a verdict that condition 1 already determines. For log duration, the registered interval is $+0.0098$ $[+0.0028, +0.0166]$, inside the ± 0.0181 region (precise null under every reading); its paired contrasts run against the observed block, two of them significantly (misalignment -0.0443 $[-0.0593, -0.0277]$; circular shift -0.0088 $[-0.0178, -0.0009]$). No within-person utility claim is licensed. (Under the superseded repetition-quantile read of Phase 12, both cells had been labeled inconclusive.) The position-curvature arm (quadratic position added to both rungs) is reported as degenerate (its Gaussian extrapolation beyond the training window is unstable), and its role is limited to documenting the mechanical basis (linear position extrapolation) that underlies these within-person gains.

Table E.1. Same-person-future M3 − M2 battery, all arms

Outcome	Arm	Reps	Mdn	$q_{.025}$	$q_{.975}$	Pr(> 0)
n:first.att	circ.shift	20	.0042	−.0042	.0702	.60
n:log.dur	circ.shift	20	.0126	.0027	.0456	1.00
n:first.att	id.swap	20	.0052	−.0025	.0876	.60
n:log.dur	id.swap	20	.0049	−.0030	.0364	.70
n:first.att	m.obs:misalign +1	20	−.0003	−.0051	.0053	.45
n:log.dur	m.obs:misalign +1	20	.0460	.0043	.1703	1.00
n:first.att	m.obs:same-item probe	20	.0113	.0014	.0868	1.00
n:log.dur	m.obs:same-item probe	20	.0049	−.0048	.0277	.90
n:first.att	misalign +1	20	−.0030	−.0090	.0029	.25
n:log.dur	misalign +1	20	−.0129	−.0669	.2571	.30
n:first.att	observed	20	.0111	.0014	.0802	1.00
n:log.dur	observed	20	.0056	.0009	.0311	1.00
n:first.att	pos.curvature	20	.0612	−.0161	1.7090	.80
n:log.dur	pos.curvature	20	−.0042	−.3860	.2955	.35
n:first.att	pseudo	20	.0038	−.0036	.1120	.60
n:log.dur	pseudo	20	.0040	−.0031	.0400	.80
n:first.att	same-item probe	20	−.0015	−.0164	.0165	.40
n:log.dur	same-item probe	20	.2621	.2428	.4021	1.00

Note. Corrected adjudication battery (Phase 12); the summaries here are repetition-level stability diagnostics (5 cuts × 4 seeds). The registered adjudication interval for the observed cells is the event-weighted participant-cluster bootstrap of the AMD-5 addendum release, and condition 3 is adjudicated on that release's paired observed-minus-control contrasts, with the scalar control summaries retained as sensitivities (Appendix E). The transport map of Table D.2 reports the earlier Phase-8 estimates, which refit identical folds under different registered tuning seeds.

E.2. **Same-person-future M4 − M3, temporal-block battery.** The original Phase-11b battery corrupted the facial base while leaving intact the temporal block, the block the contrast measures, which cannot adjudicate the contrast; the review flagged this (F-02), and Table E.2 reports the corrected battery. Both temporal-block corruptions collapse the gain to zero: within-person rotation (destroys item alignment, preserves person-level content) and cross-person swap (destroys person identity, preserves item alignment). The observed gain is therefore content-dependent (it requires the right person's temporal organization at the right items), but it is a quarter of the equivalence half-width and its repetition interval includes zero. It is recorded as a boundary of the null rather than as a finding.

TABLE E.2. Same-person-future M4 − M3 temporal battery (log duration)

Arm	Reps	Mdn	$q_{.025}$	$q_{.975}$	Pr(> 0)
observed	20	.0044	−.0014	.0109	.95
position_curvature	20	.0383	.0032	.4146	1.00
temporal_identity_swap	20	.0002	−.0047	.0210	.50
temporal_rotation_within_person	20	.0006	−.0052	.0258	.55

Note. The position-curvature arm is reported as degenerate: a quadratic position term extrapolates unstably beyond the training window (the displayed arm spans +0.0032 to +0.4146 across repetitions).

E.3. **The registered person-specific model (m5int).** The model-ladder contract specifies M5 as person-specific normalization plus person-mean × deviation interactions, asking whether the facial signal differs by person. External review (B-02) established that the person-held-out variant cannot ask that question (a held-out person's mean block is the training grand mean), so the registered specification is evaluated where it is answerable: the same-person-future regime, in which a test person's own earlier items supply the normalization history. Table E.3 reports M5int − M4 for both same-person-future outcomes. The person-held-out M5 rows in Table D.3 are reported as a redundant reparameterization stress test, per the amended contract note.

TABLE E.3. Registered person-specific interaction model (m5int): M5int − M4 in the same-person-future regime

Outcome	Reps	Mdn	$q_{.025}$	$q_{.975}$	Pr(> 0)
next_first_attempt	20	.0008	.0001	.0232	1.00
next_log_duration	20	−.0069	−.0113	.0203	.40

Note. The model-ladder contract's person-specific specification (person mean, deviation, and their interaction), evaluated in the only regime that supplies a test person's own history (train = the person's early items). Repetitions are 5 cut positions × 4 Monte Carlo seeds.

E.4. **Person-level administration-mode bootstrap.** Administration mode is constant within person, so repetition-level intervals overstate the information. Table E.4 reports person-level bootstrap intervals (n = 169; 2,000 person-cluster resamples, event-weighted per AMD-4/B-04) for the mode target alone and for the paired mode-minus-cognition differences. Every interval includes zero, which is why the main text reports the fingerprint reading as descriptive and the registered scenario is *redundancy with a descriptive fingerprint* rather than an adjudicated artifact claim.

TABLE E.4. Person-level bootstrap for the administration-mode target and paired mode-minus-cognition differences

Comparison	n	Point	$q_{.025}$	$q_{.975}$	Excl. 0
wave_alone	169	.0146	−.0312	.0599	no
wave_minus_next_first_attempt	169	.0186	−.0276	.0646	no
wave_minus_next_log_duration	169	.0184	−.0294	.0673	no
wave_minus_next_posterror_skip	168	.0341	−.0130	.0844	no
wave_minus_next_posterror_help	168	.0225	−.0286	.0706	no

Note. Event-weighted person-cluster bootstrap (AMD-4/B-04): persons are resampled as clusters and each resample statistic is the event-weighted mean, matching the registered bits-per-event estimand. The release files name the target `wave`; administration mode and wave are the same partition in these data.

E.5. **Channel-scope sensitivity.** Table E.5 compares the eight-channel primary representation with the nine-channel arm (Joy restored). Differences in medians are at most 0.0012 bits; no verdict changes. The channel-scope contract violation found by the review (F-04) therefore mattered for contract fidelity, not for conclusions.

TABLE E.5. Channel-scope sensitivity: eight-channel primary versus nine-channel arm

Outcome	Arm	Reps	Mdn	$q_{.025}$	$q_{.975}$
next_first_attempt	observed	50	−.0036	−.0062	−.0024
next_log_duration	observed	50	−.0041	−.0062	−.0021
next_posterror_help	observed	50	−.0052	−.0123	.0024
next_posterror_skip	observed	50	−.0155	−.0201	−.0093
next_first_attempt	observed_sensitivity_9ch	20	−.0043	−.0063	−.0025
next_log_duration	observed_sensitivity_9ch	20	−.0049	−.0066	−.0027
next_posterror_help	observed_sensitivity_9ch	20	−.0064	−.0119	.0020
next_posterror_skip	observed_sensitivity_9ch	20	−.0166	−.0207	−.0098

Appendix F. Adversarial Audits

The project has been audited three times at the pipeline level, and a separate editorial review examined the compiled v3 build; every round changed the record. The first audit was a *fresh-context internal AI audit*: an instance of the same model family that assisted the analysis, with no prior exposure to the project, under a written brief with six priority targets (leakage construction; end-to-end gate arithmetic; control validity; the artifact verdicts; claim boundaries; reproducibility) and a strict evidence rule: every claim in the findings had to come from files the reviewer opened or code the reviewer ran. We label it internal rather than independent because it shares model lineage with the analysis assistant; its value lay in fresh context and an explicit refutation brief, not in independent authorship. The brief, findings, and disposition logs are retained in the author-side project record. The public replication package instead ships the resulting design contracts, disclosure registers, aggregate evidence, and executable verification gates.

The review returned 15 findings (1 critical, 5 major, 6 minor, 3 notes) and an overall verdict of *sound with corrections*; it also recorded 11 attempted-and-cleared attacks. All 15 findings were accepted; none was rejected. Table F.1 summarizes.

Table F.1. First (internal) audit: findings and dispositions

ID	Severity	Finding	Disposition
F-01	Critical	Two all-positive same-person-future M3−M2 cells were computed but unreported, contradicted by synthesis-log text, and a degenerate licensing rule let single-repetition intervals license claims	Cells formally adjudicated with the full battery (Appendix E); licensing rule now requires $\geq$ 5 repetitions + adjudication survival; errata in logs
F-02	Major	The "position-trend artifact" verdict for the M4−M3 gain rested on controls that never corrupted the temporal block under test	Verdict withdrawn; temporal-block battery run; corrected reading in Appendix E
F-03	Major	M5 and the ablation leaked test persons' rows into their person means in 4 of 5 folds (fold-loop overwrite defect)	Fold-internal transform implemented; behavioral test added; Phases 8–9 rerun (direction of defect: conservative—the leaked M5 still lost)
F-04	Major	Joy sat inside every primary M3/M4 model, violating the frozen channel-eligibility contract	Eight-channel primary enforced in code; nine-channel kept as labeled sensitivity arm; full rerun; deltas $\leq$ 0.0012 bits
F-05	Major	The wave "fingerprint" headline rested on a point-median ordering whose own interval straddles zero; the promised person-level bootstrap was never run	Person bootstrap run; intervals include zero; claim downgraded to descriptive; scenario renamed
F-06	Major	The equivalence floor (AMD-2) was confirmed after provisional gate results existed, deviating from the freeze rule; a log sentence mispredicted its effect	Deviation disclosed in three artifacts; dual verdicts released; limitation stated
F-07	Minor	Bootstrap scored Gaussian cells with the marginal outcome SD, off the judgment scale; stale bootstrap columns unlabeled	Per-row own-fold training σ implemented; stale columns replaced
F-08	Minor	Controls weaker than their register text (identity swap lacked duration matching; pseudo-channel wording; coverage-only never built)	Duration-tertile swap implemented; wording aligned; coverage-only struck as degenerate (AMD-3)

ID	Severity	Finding	Disposition
F-09	Minor	Frozen interpretation gate still declared the equivalence region provisional after confirmation	Status now read from the confirmed contract
F-10	Minor	Two audit checks could never fire (`isTRUE` on a vector; a no-op structural check); fold-scope check was metadata-only	Checks fixed; fold-transform behavioral self-test added
F-11	Minor	Contract metrics (Brier/RMSE), the GBM benchmark, and the nested-CV comparison were absent without record; calibration gate used slope only	Secondary metrics released; not-run register added; intercept condition implemented
F-12	Minor	glmnet unrecorded; seeds not in releases; registry stale; manifests missing; a script was not idempotent	All recorded; manifests generated for every release; guards added
F-13	Note	"20 repetitions" in same-person-future is 5 cuts $\times$ 4 seeds	Described as such wherever cited
F-14	Note	M0's "item/position fixed effects" is a linear position index	Contract wording corrected; curvature sensitivity documented as degenerate
F-15	Note	Gaussian scorer pooled σ across folds with an all-data fallback	Row-wise fold σ implemented

F.1. **Attacks that failed.** The review also recorded what held: a constructed wave-assignment miskey (contradicted by the data), the ladder core's training-fold hygiene, the paired-rows guarantee across rungs, gate arithmetic reproduced to 10^{-10} from raw releases, transition-table time zero, retrospective-builder time zero, control mechanics spot checks, the null-distribution comparison, claim–evidence traceability, the prefacial provenance of AMD-1, and the hash-verified interface to the source releases. The defects found by this audit, in other words, concentrated in the synthesis and reporting layer, whereas the measurement layer withstood the attempted attacks.

F.2. **The second, external review.** A subsequent independent external pre-submission review of both papers in this research program audited this project again (including the closures of the first audit) and returned 25 findings across the portfolio. Its consequential Paper-B findings (each repaired and rerun; ledger in `config/amendments.yml`, AMD-4): the Phase-6 calibration had not been propagated to the eight-channel primary block; the released bootstrap weighted persons equally where the registered estimand is an event average; the gate had judged equivalence on split-stability ranges rather than the registered inferential interval; the primary retrospective cells had no matched controls and their predictors bypassed the register; the identity swap permitted fixed points; the same-person gate omitted the calibration intercept; M5 did not implement its registered interaction specification; and the clean-run driver could report success after failures. The first audit's own characterization that defects were confined to the synthesis layer was therefore too generous (both audits found implementation-level defects), and that characterization is corrected here.

F.3. **The v3 editorial review and the second external review.** Two further rounds followed the v3 restructure. First, a no-prior-context editorial review of the compiled v3 build and its sources (33 findings, 5 major) corrected reporting-layer defects: truncated evidence strings in the QA table, provenance notes for cells that two releases estimate, overstated calibration language, an invalid resolution comparison for one residual trace, unqualified freeze claims, and assorted terminology and voice issues. No separate findings artifact was retained; the applied fixes are itemized in the manuscript's change log. Second, an independent external review of the assembled v3 build (the second external review of this project, and the third pipeline-level audit) returned six must-fix findings and led to AMD-5 (Appendix B): the facial magnitude summary had been described as a peak where the implemented feature is a conditional mean log magnitude (description corrected; nothing refit); the within-person cells had been adjudicated on repetition quantiles where the contract requires the participant-cluster bootstrap (the registered object is now supplied by the addendum release, Appendix E; the duration cell became a precise null, and the first-attempt cell became the study's one comparator-dependent verdict, reported under both control summaries); a sensitivity ratio had divided a Gaussian artifact by the binary-only resolution anchor (replaced by family-valid comparators); the mixed-mode evidence basis, the published Zoom-only record, and the undocumented in-person capture path are now stated explicitly; headline prose was narrowed to the released verdict categories; and the signed *Negative* channel's provenance was corrected. That review also verified the released arithmetic independently, refitting one primary cell, the same-item probe, and the administration-mode benchmark exactly.

A focused re-review by the same external reviewer then audited the AMD-5 changes. It reproduced the addendum release byte for byte from an isolated copy, confirmed the corrected descriptions and the primary results, and returned two must-fix findings on the changes themselves: neither scalar summary of the control arms implements the contrast the registered reporting minimum names, and the mechanism then attributed to the pseudo-channel arm (supplying item-position information) is not established, because that arm preserves no item structure. Both were accepted. The addendum release now also refits the four corrupted arms under the identical seeds and adjudicates condition 3 on the paired person-cluster contrasts, with the two scalar summaries retained as disclosed sensitivities and the first-attempt cell's pass under the historical median rule stated wherever the cell is reported; the mechanism prose was narrowed to what the identity swap supports; and the semantic correction was completed in the frozen ladder contract itself.

F.4. **Effect on conclusions.** After all corrections and reruns from every round: the 0-of-6 gate outcome, the negative direction of every primary observed contrast, and the M2 frontier minimum are unchanged. Under AMD-5 the within-person duration cell became a precise null, and the first-attempt cell became the one cell whose control comparison carries weight: it passes the historical scalar rule, and the paired contrasts that the reporting minimum registers resolve it as not separated from its controls, so it licenses no claim. No verdict anywhere moved toward a facial-utility claim. What changed is recorded: the scenario was downgraded from an adjudicated artifact-plus-redundancy reading to *redundancy with a descriptive fingerprint*; the within-person cells moved from unreported to adjudicated-and-unlicensed; the equivalence comparator, the inferential interval, and the control coverage were each replaced by their contract-faithful versions; and the personalization question was re-posed in the regime that can answer it.

Appendix G. **Statistical Quality Assurance and Reproducibility**

G.1. **Validation checks.** Table G.1 reproduces the programmatic validation checks frozen with the evidence release. The register's governing rule applies throughout: an analysis that fails an automated check is invalid regardless of its result.

TABLE G.1. Statistical QA checks

Check	Status	Evidence
primary_cells	PASS	6 primary cells
controls_present	PASS	5 control arms with matched observed arms

Check	Status	Evidence
ablations_present	PASS	2 ablation arms (fold-internal person means, F-03 fix)
folds_shared	PASS	all rungs read config/folds/*.csv
delta_status	CONFIRMED	equivalence-region-v3.json status: confirmed_by_user (D3, AMD-2)
null_distribution_negative	PASS	null q975 all < 0 (max single replicate 0.00020; A1 anchor uses q975)
frontier_minimum_rung	PASS	M2 has the lowest median codelength for every outcome
scenario_recorded	PASS	redundancy_with_descriptive_fingerprint: no representation passed Gate B in any primary population-transport cell and the within-person positive cells did not separate from corrupted-facial controls or fell below the equivalence threshold; the same representation predicted recording wave better than any cognitive outcome as a point median, but the person-level wave interval includes zero, so the fingerprint reading is descriptive
adjudication_present	PASS	adjudication_v1 release joined into Gate C and the scenario
channel_scope_primary	PASS	primary M3 excludes Joy (AMD-3); nine-channel arm is sensitivity-only
gate_c_min_reps	PASS	licensing requires >= 5 repetitions and adjudication survival
calibration_scope_matches_primary	PASS	phase06 null calibrated on the 24-column primary block (B-01); 27-column scope labelled sensitivity
inferential_interval_event_weighted	PASS	participant-cluster bootstrap resamples persons, event-weighted mean (B-04); gate condition 1 uses it (B-05)
retro_controls_present	PASS	8 retrospective control cells in retro_block_12 (B-06)

G.2. **Environment and invocation.** R 4.6.0 (aarch64-apple-darwin23), glmnet 5.0 (the fitting engine of every rung), testthat 3.3.2 for the register and fold-transform self-tests; figures with ggplot2 and patchwork; documents with TeX Live

2025. All scripts are invoked as RENV_CONFIG_AUTOLOADER_ENABLED=FALSERscript--vanilla<script> from the codebase root. The public config/folds/_SEEDS.json records fold architecture and counts but not the empirical seed; that seed and the empirical assignments remain custodian-only. Public non-empirical phase seed bases remain recorded in the corresponding release _SUCCESS.json.

G.3. **Release interface.** Every analysis statistic in the article traces to a release directory with a _SUCCESS.json sidecar and a SHA-256 checksum-manifest.csv: baseline_v1, phase06-simulation_v1, m3_v1, m4m5_v1, transport_v1, adjudication_v1, evidence_v1 (the gate, claims, and canonical uncertainty tables), statistical-qa_v1, and the AMD-5 addendum spf-person-bootstrap_v1 (registered within-person intervals, paired control contrasts, and both scalar comparator sensitivities), whose refits, observed and corrupted arms alike, are fidelity-gated against adjudication_v1 before release. Source-study assets are consumed read-only through a hash-verifying accessor; nothing writes into the source tree, and no source-study statistic is re-estimated. The archival mode descriptives of Appendix A are emitted by a guarded custodian-only producer from the immutable raw stores and are not part of the frozen releases. In the public package, root discovery is marker-based and its manifest-only cold-start rehearsal exercises both absolute and relative invocation from relocated paths containing a space.

G.4. **Number verification.** The author-side manuscript build contains a verification script, notes/verify-numbers.py, which compares an enumerated set of statistics printed in the main text and this supplement (currently well over one hundred checks) against the release files, or, for the archival corpus descriptives, against the build-time extract of Appendix A, and halts with an explicit error on any discrepancy; the submitted build requires an ALL PASS run. Two limits of this gate should be stated. It checks enumerated literals and invariants rather than parsing the rendered manuscript, so a printed statistic outside its enumeration is unprotected; and for the archival descriptives it compares the prose against a cache written by the same extraction script, so it catches drift after extraction but cannot catch an error inside the extraction logic itself. The public package adds a distinct typed audit of 25 selected printed values, each with a TeX locator and hashed evidence excerpt. Phase-by-phase execution logs remain in the author-side project record; the public package carries the resulting provenance registers and structured live release status.